

DESIGN
CONSTRUCT
OPERATE

PRIMERO

PROCESS DESIGN CRITERIA - STAGE 2

KCGM GROWTH PROJECT

2000-F00-DCR-PR-10002

NORTHERN STAR RESOURCES LTD



PRIMERO PROJECT NUMBER: 23517

DATE	REVISION	STATUS	PREPARED	REVIEWED	APPROVED
7-Jun-24	0	Issued for Design			
18-Oct-24	1	Reissued for Design			
10-Dec-24	2	Reissued for Design			
14-Feb-25	3	Reissued for Design			
22-Jul-25	4	Updates to BF Loaded Carbon Circuit and Cyanide Concentration			

PROCESS DESIGN CRITERIA

NORTHERN STAR RESOURCES LTD
KCGM GROWTH PROJECT
2000-F00-DCR-PR-10002

DISCLAIMERS

This file has been prepared for Northern Star Resources Ltd (NSR) by Primero Group Pty Ltd (Primero) as an independent consultant and is based in part on information furnished by NSR and in part on information not within the control of either NSR or Primero. While it is believed that the information, conclusions and recommendations will be reliable under the conditions and subject to the limitations set forward herein, Primero does not guarantee their accuracy. The use of this report and the information contained herein shall be at the user's sole risk, regardless of any fault or negligence of Primero.

The information, formulae and calculations presented in this spreadsheet represent significant intellectual property and effort, are provided in good faith, and under the understanding that NSR will not provide the working version of this spreadsheet to any other companies without Primero's express permission.

REFERENCES

Client	NORTHERN STAR RESOURCES LTD
Project	KCGM GROWTH PROJECT
Document	PROCESS DESIGN CRITERIA - STAGE 2

No	Document	Description
1	5.2.2.1 - Site Specific Information_Rev 1	KCGM Growth Project - Process Plant - Site Specific Information
2	4.7.1 - OMC Report 7555 - KCGM PDC Data Summary Rev 0, March 2022	Table 2-5, "Back Calculated Ore Parameters", 2014 - 2021 value.
4	4.6.1 - KCGM Growth Mass Balance_Nominal Case Report_Rev C	Mass Balance - SysCAD Report Nominal Case. 22/03/2022
5	"KCGM Comminution Plant Data_B.xlsx"	Benchmarking from Fimiston pebble crushing circuit.
7	ECI Tender from Concep	Q12375 - Primero Gravity Circuit Rev1
8	5.1.12f - BOD Alignment Template - Flotation Tail PLT, FT CIL, Elution & EW	Basis of Design for Flotation Tailings CIL, Elution & EW circuit. 06/04/2022
9	4.7.3 - OMC Report 7555 KCGM PDC PSD Addendum Rev 1	KCGM - Process Design Criteria Summary - PSD Addendum Report No. 7555 Rev 1 March 2022
10	5.1.12g - BOD Alignment Template - Reagents and Services	5.1.12g - BOD Alignment Template - Reagents and Services - Rev C NSR confirmation that regrind mill ball size will be 25 mm (cell F15).
11	Primero Calculation 23517-CAL-PR-002	Flotation Tailings CIL Simulation @ 8 and 10 g/L Carbon Concentration
14	KCGM Growth Project ECI Outcome Letter - Primero	KCGM Growth Project - Process Plant - Formal Agreement for Engineering, Procurement and Construction: Early Contractor Involvement (Value Engineering Phase). 03/08/2022
15	5.1.12b - BOD Alignment Template - 13 Grinding - Rev C	5.1.12b - BOD Alignment Template -13 Grinding - Rev C - Primero (220609)_NSR Review
16	KCGM Process Design Criteria - VEP Phase (Stage 1)_I	KCGM - Growth Project - Process Design Criteria Rev I. 20/09/2022
18	Cleaner Option 2- KCGM Case 2 Scav Con to Cleaner	Glencore Technology - Jameson Cell Mass Balance - Cleaner
19	Cleaner Scavenger - KCGM Case 2 Scav to Cleaner	Glencore Technology - Jameson Cell Mass Balance - Cleaner Scavenger
23	KCGM GP Process Design Criteria - Reliance Information S1 Rev K, 25/01/2023	Principal supplied reliance information.
24	VEP Outcome Letters - Primero, 09/02/23 and 020523	Principal supplied Scope of Works letter for Stage 1 and 2
25	OMC Report 7653 - KCGM Value Engineering Update	Grinding circuit modelling for KCGM by Orway Mineral Consultants, 160223
26	Process Design Workshop Meeting Minutes 170223, 23517-MOM-PM-SFP002	Workshop between the Client and Primero
27	Glencore Jameson Cell Sizings and Quote	Simulations from Vendor, 22/02/24 KCGM Expansion Project - Jameson Cell Quote OTJ-2220 7532.3
28	Weir Cyclone Simulations and Datasheet - Primary & Gravity Cyclones	Various simulations for the different process cases, and datasheet: 2000-F00-DSH-ME-11005A_B Primary Cyclone
31	Trash Screen Spray Water Information from Vendor	Emails with Vendor, 16/08/23
32	Flotation Mass Balance Scenarios	Mass Balances for Process Margins Rev H and New Mass Balance Cases Rev G
33	Cadia Gravity Electrowinning Cell	Emails with Vendor, 14/03/23. Ag and Cu Assays from site, provided by NSR (23/03/23)
35	Discussions with Client and TQ regarding Cyanide Destruction	Emails with Client, 230323 and TQ: 2000-F72-TQY-ME-10094_0_TQ-R
36	Vendor Information from Transmin for New Flocculant Mixing Plant	Transmin Proposal Execution Phase rev 3, P33-904-06
37	Direction from Primero Mechanical Engineers re: COS	Email, 280323
38	Client Information regarding Grinding Media	Emails with Client, 070323
39	Vendor Information from Concep for the Acacia CS8000 ILR	Emails with Client, 210322
40	Primero Calculation	"SAG_Ball_Screen_Trommel_Calc_Rev_A.xlsx"
41	Client Direction, Transfer of FLS Equipment Technical Information	Emails from client, various dates (May 2023)
42	Primary Crusher Datasheet	Scope of Work - Primary Crusher, Appendix A, doc 2000-F12-SOW-ME-20001 rev 1
43	SoW for Separable Portions Stage 1&2	Scope of Works between NSR and Primero: 2000-F00-SOW-CO-20001 & 2000-F00-SOW-CO-20002_0
44	Outstanding Process Inputs from NSR	Outstanding Process Inputs 210623 (version 1) Additional (003) (002), and 23517-2000-F00-TQY-PR-10051.0.TQ-R.0.01
47	Oxygen Consumption Information	Emails from client, 8/5/23
48	Cadia CIL4 Electrowinning Cells	Vendor Quotation 27/03/2024
49	NSR email "KCGM GP - Float Cell Motors", dated 11 May 2023	FLSmidth flotation blower details.
50	Calculation for LPG Consumption and TPE ECI Tender	23517-CAL-PR-005_A LPG Consumption TPE5000 ELUTION HEATER PACKAGE - 23517-ME16 TENDER - COMPLETE 20/05/22
51	FLSmidth - Primary Crusher Application Model	FLS_13x19 TSUV_Calc_Fimiston_Run 15_13x19 TSUV_6.5 OSS_3716 mtph_70, AVG results
52	Vendor Information Mercury Abatement System (CIL4 Carbon Regen Kiln) and Client Direction Variation 25; Vendor information for Lochhead CIL4 Carbon Regen Kiln; Regen Kiln Off Gas Calculations	Information from the Mercury Abatement Vendor, Jan 2024 KCGM-P3-EPC-COR-161 Variation Direction No.025 Carbon regeneration kiln off-gas mercury abatement system Lochhead Carbon Reactivation Kiln (ME-17) Quotation No. 5211 (August 26, 2022) Kiln Off Gas Calc
53	Primero email "23517 Apron Feeder Datasheet Internal Review", dated 30 August 2023	Agreement with NSR to change contract SOW design factor from 120% to 110% of nominal tonnage rate.
54	Weir UFG Cyclone Cluster Modelling	UFG_KCGM_CY_Select_Design_RevC UFG_KCGM_CY_Select_Nominal_RevC
55	Takraf Tender Rev 0 Datasheet	2000-F00-DSH-ME-11018_B Trash Screens Completed
56	KCGM GP Process Design Criteria - Reliance Information S1 Rev L, 25/01/2023	Principal supplied reliance information.
57	KCGM Growth Project SysCAD Model (Historical)	SysCAD Model files and Reports (2022.08.31)-01.spf
58	NTF Deslime & UFG Cyclone PDC Inputs	Word document 260723
59	Joest Vibrating Screens Offer	JOEST_M22-11102 Rev D_JOEST FIRM Offer 2000-F00-DSH-ME-11011_C Vibrating Screens
60	Flotation Launder Wash Water Technical Query	Technical Query (2000-F16-TQY-ME-10007) FLS Launder Water Requirements Rev.0 31/07/2023
62	Technical Query - Design Bulk Density Range for Apron Feeders And Conveyors	2000-F00-TQY-ME-10034

REFERENCES

Client	NORTHERN STAR RESOURCES LTD
Project	KCGM GROWTH PROJECT
Document	PROCESS DESIGN CRITERIA - STAGE 2

No	Document	Description
63	Flotation Area - Design P80 Sizes	Email Correspondence. RE 23517 - Flotation D80 Mech Requests. 250823
64	Reagent Delivery Formats, 2000-F72-TQY-ME-20001_2	Reagent delivery information, provided by NSR.
65	IsaMill Quote and Modelling by Glencore	IsaMill Quote OMT232007747 KCGM - Northern Star (Primeo) - IsaMill Sizing - Design -19102023 - New data supplied water temp 25 KCGM - Northern Star (Primeo) - IsaMill Sizing - Nominal -19102023 - New data supplied water temp 25 Email: "FW_ ME039 IsaMill Latest MB - Clarifications 64 - UFG Mill Media Hopper Live Volume"
66	Regrind Removal TQ	2000-F00-TQY-GE-10080_0 - RESPONSE.pdf
67	Client Comments Stage 1 Rev K PDC and Rev L SysCAD	2000-F00-DCR-PR-10001.K.ICR-R.K.02
68	Raw Water TQ 10044	23517-2000-F24-TQY-PR-10044.0.TQ-R.0.0164
69	SysCAD & PDC Client Comments TQ 10107 Response	2000-F00-TQY-PR-10107.0.TQ-R.0.02
70	Gravity Variation (VAR-017) and Communication from the Client	KCGM-P3-EPC-COR-085 Variation Direction No.017 Gravity Circuit Modifications Emails from Client/Outcome of meetings
71	Emails from Concep (Knelson & ILR) in relation to Gravity Variation (VAR-017)	Email dated 30/10/23, 'RE: Knelson Concentrators (KC-QS70) and CS10000'
72	Gravity Variation TQ 10115 Response from Client	2000-F14-TQY-PR-10115.0.TQ-R.0.01 2000-F14-TQY-PR-10115_1_TQ-R
73	Quotation and emails from Cadia Systems (EW cells) in relation to Gravity Variation (VAR-017)	Q231115-PRI_PRIMERO_EWC 151123 issue RE_ 23517 KCGM - Gravity Pregnant Solution Electrowinning - various emails
74	Variation 018 UFG Mill 2 Trash Screen	KCGM-P3-EPC-COR-104 Variation Direction No.018 UFG Mill 2 Trash Screen
75	Clarification Emails for Flotation Reagent Calculations	Emails with Client 21/11/23: 'RE: 23517 KCGM GP - PAX Usage/Flow Consumption/Storage Tank RT'
76	Dump Pocket TQ from NSR	2000-F12-TQY-GE-10127_0_TQ-R
77	Wet Static Screens Datasheet (ME40) and Spray Water Information	2000-F00-DSH-ME-11040_1 and emails dated 23/01/24
78	NSR Owner's Manual CAT793F KCGM	Owner's Manual CAT793F KCGM
79	Water Services PFD Client Review	23517-2000-F00-PFD-PR-10046.D.IFD-R.D.02
80	Area 30 M1 Design Review Report	2000-F30-REP-GE-10101_A - M1 Design Review Report
81	TPE CIL4 Elution Heater Vendor Package	TPE5000 ELUTION HEATER PACKAGE_DES.pdf rev A 20/05/22
82	Client Information LPG Storage Vessel	Email: RE 23517 LPG Tank, 22/05/2023
83	TQ147 Reagent Tank Volumes, Var 26 Cyanide Tanks and Var 192 CuSO4 Tanks	2000-F72-TQY-ME-10147_0_TQ-R KCGM-P3-EPC-COR-162 Variation Direction No.026 Cyanide Tank Storage Capacity KCGM-P3-EPC-COR-192 Variation Direction No.030 Copper Sulphate Storage
84	Joest Tender Documents	JOEST_M22-11102 Rev H_JOEST FIRM Offer, 06/05/24 RE_KCGM Growth Project - Stage 2 Vibrating Screen Changes
85	Gekko Carbon Scout ECI Tender Documents	6454-Primero-KCGM-Carbon Scout-220426, 22/04/22
86	ME65 Carbon Transfer Pump Vendor Information	Weir Budget Quote WMFN20220527
87	ME07 Intertank Screens Vendor Information	P28337_Northern Star_Primero Group_KCGM Growth (Aus) REV 2
88	TQ206 Request for LPG & RTO Information	23517-2000-F00-TQY-PR-10206.0.TQ-R.0.0270
89	2000-F00-CAL-ME-F23TK404 Final Tails Tank Live Vol & OF Calc.xlsx	2000-F00-CAL-ME-F23TK404 Final Tails Tank Live Vol & OF Calc.xlsx
90	Client Comments Stage 1 Rev 0 PDC	23517-2000-F00-DCR-PR-10001.0.IFD-R.0.02
91	Stage 2 Oxygen Consumption Calculations	23517-CAL-PR-003 S2 Oxygen Consumption Calculations
92	SAG & BM Trommel Sizes	2000-ME001.2-B02-50041_C SAG Mill Trommel Sizing Report 2000-ME001.2-M03-50001_B SAG Mill Trommel Assembly 2000-ME001.3-B02-50020_B Ball Mill Trommel Sizing Report 2000-ME001.3-M03-50001_B Ball Mill Trommel Assembly
100	5.1.1 - ECS Process Design Criteria	DRA Process Design Criteria - KCGM Growth Project ECS. 25/01/2022 J6027-000-DCR-PR-001 Rev D
101	5.1.2 - ECS Mass Balance	DRA Mass Balance - KCGM Growth Project ECS. 25/01/2022 J6027-000-CAL-PR-001 Rev D
102	4.5.1 - Process Design Criteria Rev F	KCGM - Growth Project - Process Design Criteria Rev F. 16/03/2022
103	4.5.1 - KCGM Process Design Criteria - VEP Phase _H, 19/08/2022	Principal supplied reliance information.
104	Email "Carbon Safety Screen" dated 16/02/2024	Confirmation from NSR that currently proposed screen sizes for Carbon Safety Screens is adequate
105	2000-F30-TQY-PR-15001.0.TQ-R.0.02	Client Response to Technical Query - CIL2/3 - Carbon Transfer Rate Between Adsorption Tanks
106	2000-F00-DCR-PR-10002.C.ICR.-R.C.02	Client comments to Process Design Criteria - Stage 2
107	2000-F72-TQY-ME-10260_0-R	Clients Response to Technical Query - Milk of Lime Tank Volume & Residence Time
108	2000-F21-TQY-PR-15014.0.TQ-R.0.02	Clients Response to Technical Query - CIL2/3 & CIL4 Target Process Data
109	2000-F00-TQY-PR-15020.0.TQ-R.0.02	Clients Response to Technical Query - CIL4 Carbon Advance
110	2000-F22-TQY-PR-15018_1_RSP	Clients Response to Technical Query - HCI Dilution Tank
111	Hydrogen Peroxide Technical Data Sheet	Solvay Hydrogen Peroxide Technical Datasheet

REFERENCES

Client	NORTHERN STAR RESOURCES LTD
Project	KCGM GROWTH PROJECT
Document	PROCESS DESIGN CRITERIA - STAGE 2

No	Document	Description
112	PAX, CuSO4 & Lead Nitrate Delivery Sizes	Email from IXOM (vendor), dated 28/03/2024
113	Concentrate Thickener Vendor GA	2000-ME091-B02-10001_C
114	Client Comments Stage 2 Rev 0 PDC	20000-F00-DCR-PR-10002_0_AAN
115	2000-F30-TQY-PR-15029_0_TQ-R	Clients Response to Technical Query - Concentrate Pre-Leach Thickener Underflow
116	2000-F72-TQY-PR-15033_0_TQ-R	Clients Response to Technical Query - Lime System Process Design
117	QUO-07322-P7S5G0	Derrick quotation for Derrick Linear Motion Screening Machine 15/08/2024
118	BHT Lime Slaking Package	2000-F00-DSH-ME-12013_0 2000-MEP-ME013-REG_A 706-PC-01 706-PC-02 FW_Lime Mill Media Consumption.msg
119	2000-F30-TQY-PR-15097_1	CIL3 Carbon Movement TQ

AREA	PARAMETER	UNIT	DATA					REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV
				1	2	3	4					
			FIMISTON	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um	NEW CIRCUIT 28.35 MTPA THROUGHPUT NOMINAL GRADE P80 220um					
1.0	PRODUCTION SCHEDULE											
1.1	Throughput											
	Annual Throughput	t/a	10,500,000	25,500,000	23,997,254	25,500,000	28,350,000	1	Northern Star	16, 24, 26 & 32		A
	Annual Operating Days	d/a	365	365	365	365	365	1	Northern Star	16		A
1.2	Operating Schedules											
	Crushing											
	Plant Availability	%	65.0	65.0	65.0	65.0	65.0	1	Northern Star	1, 30		A
	Annual Operating Hours	h/a	5,694	5,694	5,694	5,694	5,694	6	Calculated			
	Hourly Throughput - Existing Circuit	th	1,844	1,844	1,844	1,844	1,844	6	Calculated			
	Hourly Throughput - New Circuit	th		2,634	2,370	2,634	3,135	6	Calculated			
	Total Hourly Throughput	th	1,844	4,478	4,214	4,478	4,979	6	Calculated			
1.3	Concentrator											
	Plant Availability	%	93.3	93.4	93.4	93.4	93.4	1	Northern Star	16, 23		A
	Annual Operating Hours - Existing SAG Circuit		8,000	-	-	-	-	6	Calculated			
	Annual Operating Hours - New SAG/Ball Mill Circuit	h/a	-	8,182	8,182	8,182	8,182	6	Calculated			
	Hourly Throughput - Existing SAG Circuit	th	1,313	1,313	1,313	1,313	1,313	6	Calculated			
	Hourly Throughput - New SAG/Ball Mill Circuit	th		1,833	1,650	1,833	2,182	6	Calculated			
	Total Hourly Throughput	th	1,313	3,146	2,962	3,146	3,494	6	Calculated			
	UFG Circuit											
	Plant Availability	%	93.3	93.4	93.4	93.4	93.4	1	Northern Star	16, 23		A
	Annual Operating Hours	h/a	8,000	8,182	8,182	8,182	8,182	6	Calculated			
2.0	MATERIAL CHARACTERISTICS											
2.1	General Ore Properties											
	Moisture Content	Nominal	%w/w		1.3			1	Northern Star	16		A
	Ore Specific Gravity	Range			2.90			1	Northern Star	16		A
					2.70 - 3.10			1	Northern Star	23		B
	Ore Bulk Density	Normal			1.90			1	Northern Star	16		A
		Range			1.70 - 2.10			1	Northern Star	56		B
	Angle of Repose	Design Range			1.60 - 1.90			1	Northern Star	62	To be used for all conveyors & apron feeders in the crushing and milling circuits	B
	Drawdown Angle		degrees		36			1	Northern Star	16		A
			degrees		60			1	Northern Star	16		A
2.2	Ore Comminution Parameters											
	Crushing Work Index											
	Testwork Average		kWh/t		17.8			1	Northern Star	16		A
	Testwork Range				16 - 30			1	Northern Star	16		A
	Testwork 80th Percentile		kWh/t		18.8			1	Northern Star	2, 23	2014 - 2021 value (as of rev E)	A
	Abrasion Index											
	Testwork Average				0.145			1	Northern Star	16		A
	Testwork Range				0.020 - 0.680			1	Northern Star	16		A
	Testwork 80th Percentile				0.192			1	Northern Star	16		A
	Ball Mill Work Index	Range	kWh/t		11.5 - 15			1	Northern Star	41		B
	A*b	Design Range	kWh/t		12.5			1	Northern Star	2	2014 - 2021 value	A
					38 - 48			1	Northern Star	41		B
					43.8			1	Northern Star	2	2014 - 2021 value	A
2.3	Pebble Comminution Parameters											
	Crushing Work Index	Design	kWh/t		18.8			1	Northern Star	16		A
		Maximum	kWh/t		23.0			1	Northern Star	23		A
	Bulk Density	Minimum	tm ³		1.60			1	Northern Star	16, 62		A
		Maximum	tm ³		1.90			1	Northern Star	41, 62		B
2.4	Non Ore Grades											
	Feed Grade - S	% S		1.02	1.02	1.40	1.02	1	Northern Star	23	Achievable grade back calculated by Primoero (Source 32)	A
	Feed Grade - Au	Au (g/t)		1.58	1.58	1.70	1.58	1	Northern Star	23	Achievable grade back calculated by Primoero (Source 32)	A
	Final Tail Grade - S	%		0.18	1.01	1.39	1.01	1	Northern Star	23	Blue data is from NSR Ref 23	A
	Final Tail Grade - Au	Au (g/t)		0.23	0.23	0.25	0.23	1	Northern Star	23	Blue data is from NSR Ref 23	A
2.5	Design Ore Grades											
	Feed Grade - S	%S		1.49	1.49			1	Northern Star	23		A
	Feed Grade - Au	Au (g/t)		1.81	1.81			1	Northern Star	23		A
	Final Tail Grade - S	%		0.29	1.48			1	Northern Star	23	Blue data is from NSR Ref 23	A
	Final Tail Grade - Au	Au (g/t)		0.38	0.26			1	Northern Star	23	Blue data is from NSR Ref 23	A
2.6	Recovery											
	Total Gold Recovery	Nominal	%	84	84	84	84	6	Calculated			
	Gravity Gold Recovery	Nominal	%	14.00	14.00	14.00	14.00	1	Northern Star	23	Relative to feed grade	A
	Gravity Gold Recovery	Maximum	%	20.00	20.00	20.00	20.00	1	Northern Star	23	Relative to feed grade	A
	Flotation Gold Recovery	Nominal	%	71.80	71.80	71.80	71.80	1	Northern Star	23	Relative to feed grade	A
	Flotation Gold Recovery	Maximum	%	75.00	75.00	75.00	75.00	1	Northern Star	23	Relative to feed grade	A
	Flotation Sulphur Recovery	Nominal	%	85.10	85.10	85.10	85.10	1	Northern Star	23	Relative to feed grade	A
	Flotation Sulphur Recovery	Maximum	%	90.00	90.00	90.00	90.00	1	Northern Star	23	Relative to feed grade	A
	Flotation Concentrate Leach Gold Recovery	Nominal	%	84.00	84.00	84.00	84.00	1	Northern Star	44	Range 80-88%	A
	Flotation Concentrate Leach Gold Recovery	Maximum	%	88.00	88.00	88.00	88.00	1	Northern Star	44	Range 80-88%	A
	Flotation Tails Leach Gold Recovery	Nominal	%	34.00	34.00	34.00	34.00	1	Northern Star	44	34% YTD	A
	Flotation Tails Leach Gold Recovery	Maximum	%	34.00	34.00	34.00	34.00	1	Northern Star	44	34% YTD	A
2.7	Gold Production											
	Nominal	oz Au/a		1,108,133	1,194,627	1,195,298	1,231,759	6	Calculated			
2.8	Flotation Concentrate Production											
		t/a		636,932	675,971	676,099	707,270	6	Calculated			
2.9	Process Water SG											
				1.070	1.070	1.070	1.070	1	Northern Star	1		A
2.10	Scheme Water SG											
				1.000	1.000	1.000	1.000	1	Northern Star	1		B
3.0	CRUSHING CIRCUIT											
3.1	Circuit Parameters											
	Throughput	Nominal	th	1,844	2,634	2,370	2,634	6	Calculated		New circuit	
		Design	th		3,135	3,135	3,135	1	Northern Star	43		
		Maximum	th		3,436	3,436	3,436	1	Northern Star	56	Crusher fall through from FLS and not the circuit design basis	B
	Feed PSD											
	F ₁₀	mm		1,000	1,000	1,000	1,000	1	Northern Star	23		A
	F ₅₀	mm		450	450	450	450	1	Northern Star	23		A
	F ₈₀	mm		150	150	150	150	1	Northern Star	23		A
	F ₂₀₀	mm		25	25	25	25	1	Northern Star	23		A
	Product PSD											
	P ₈₀ - Maximum	mm		100	130	130	145	1, 5	Northern Star, Vendor	23, 25		A
3.2	Dump Pocket											
	Dump Pocket Type			Direct Tip, Two Sides	Direct Tip, Two Sides	Direct Tip, Two Sides	Direct Tip, Two Sides	1	Northern Star	16		A
	Truck Model			793F	793F	793F	793F	1	Northern Star	23		A
	Truck Payload	t		250	250	250	250	1	Northern Star	78	Owner's Manual CAT793F.	B
	SAE 2:1 Heaped Tray Capacity	m ³		179	179	179	179	1	Northern Star	76, 78	Owner's Manual CAT793F.	B
	Dump Pocket Capacity	m ³		179	179	179	179	1	Northern Star	76, 78	Live capacity from one side	B
3.3	Primary Crusher											
	Type			Gyratory	Gyratory	Gyratory	Gyratory	1	Northern Star	16	NSR Supplied	A
	Model			54-75Mk II	1300 x 1900 TSUV	1300 x 1900 TSUV	1300 x 1900 TSUV	1	Northern Star	41	NSR Supplied	A
	Open Side Setting	mm		141	165	165	165	5	Vendor	51		B
	Throw	mm		38	38	38	38	1	Northern Star	42		B
	Motor Power	kW		450	600	600	600	5	Vendor	42		A
	Simulation Results											
	P ₅₀	Max	mm		122	122	122	5	Vendor	51	Simulated, see reference document for inputs	B
	P ₁₀₀	Max	mm		229	229	229	5	Vendor	51	Simulated, see reference document for inputs	B

AREA	PARAMETER	UNIT	DATA					REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV
			1	2	3	4						
			FIMISTON	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um	NEW CIRCUIT 28.35 MTPA THROUGHPUT NOMINAL GRADE P80 220um					
3.4	Crushed Ore Storage Type		Stockpile	Stockpile	Stockpile	Stockpile	Stockpile	1	Northern Star	16		A
	Live Capacity - Existing	t	21,001	-	-	-	-	6	Calculated			1
	Residence Time - Existing	h	16	-	-	-	-	6	Calculated			
	Bulk Density	tm ³		1.60 - 1.80	1.60 - 1.80	1.60 - 1.80	1.60 - 1.80	1	Northern Star	43		B
	Live Capacity - New	m ³		12,035	12,035	12,035	12,035	2	Primero	37		A
	Residence Time - Minimum/Range	h		11.16	12.40	11.16	9.38	6	Calculated			
	Live Capacity at 1.70 tm ³	t		20,460	20,460	20,460	20,460	6	Calculated			
	Live Capacity at 1.90 tm ³	t		22,867	22,867	22,867	22,867	6	Calculated			
	Live Capacity - New	m ³		11,293	11,293	11,293	11,293	2	Primero	37		A
	Residence Time - Minimum/Range	h		10.47	11.64	10.47	8.80	6	Calculated			
	Live Capacity at 1.70 tm ³	t		19,198	19,198	19,198	19,198	6	Calculated			
	Live Capacity at 1.90 tm ³	t		21,457	21,457	21,457	21,457	6	Calculated			
	Primary Crusher Discharge Feeder Number		1	1	1	1	1	1	Northern Star	43		B
	Type		Apron	Apron	Apron	Apron	Apron	1	Northern Star	16		B
	Capacity	Nominal Design	t/h t/h	2,634 3,135	2,370 3,135	2,634 3,135	3,135 3,135	6	Calculated			
								1	Northern Star	43		B
3.5	Reclaim Feeders Number		4	2	2	2	2	1, 2	Primero, Northern Star	16, 26		A
	Type		Apron	Apron	Apron	Apron	Apron	1	Northern Star	16		A
	Capacity per feeder	Nominal Design	t/h t/h	917 2,017	917 2,017	917 2,017	1,091 2,017	6	Calculated			
								9	Agreed	53	110% of nominal tonnage rate.	B
3.6	Emergency Feeders Number		1	1	1	1	1	1	Northern Star	16		A
	Type		Apron	Apron	Apron	Apron	Apron	1	Northern Star	16		A
	Capacity per feeder	Nominal Design	t/h t/h	1,833 2,017	1,833 2,017	1,833 2,017	2,182 2,017	6	Calculated			
								9	Agreed	53	110% of nominal tonnage rate.	B
	Loader Model			992K/994K	992K/994K	992K/994K	992K/994K	1	Northern Star	23		A
4.0	GRINDING AND CLASSIFICATION											
4.1	Circuit Parameters Type		SABC	SABC	SABC	SABC	SABC	1	Northern Star	16		A
	Nom Throughput	Nominal @ 180 µm	1,313	1,833	1,650	1,833		6	Calculated			
	Nom Throughput	Nominal @ 220 µm					2,182	6	Calculated			
	Bulk Density	Design Range		1.60 - 1.90	1.60 - 1.90	1.60 - 1.90	1.60 - 1.90	1	Northern Star	43, 62		B
	Milling Circuit PSD							1, 5	Northern Star, Vendor	23, 25		A
	P ₈₀	Target	100	130	130	130	145	1	Northern Star	16, 23, 24		A
	P ₈₀		180	180	180	180	220					
	SAG Mill Specific Energy - 50th Percentile		8.3					1	Northern Star	16		A
	SAG Mill Specific Energy - 80th Percentile		8.9					1	Northern Star	16		A
	Ball Mill Specific Energy - 50th Percentile		6.4					1	Northern Star	16		A
	Ball Mill Specific Energy - 80th Percentile		6.1					1	Northern Star	16		A
	P ₈₀	Modelled		189	189	189	220	1	Northern Star	25		A
	SAG Mill Specific Energy Modelled		9.14	9.14	9.14	7.68		1	Northern Star	25		A
	Ball Mill Specific Energy Modelled	@ 180 µm	9.68	9.68	9.68			1	Northern Star	25		A
	Ball Mill Specific Energy Modelled	@ 220 µm				8.14		2	Primero	25		A
4.2	Primary Mill Type		SAG Mill	SAG Mill	SAG Mill	SAG Mill	SAG Mill	1	Northern Star	16	NSR Supplied	A
	Mill Diameter		10.97	10.97	10.97	10.97		1	Northern Star	16	NSR Supplied	A
	Effective Grinding Length - EGL		4.27	8.38	8.38	8.38		1	Northern Star	16, 23	NSR Supplied	A
	Duty Mill Speed Range	% Cs	60 - 80	60 - 83	60 - 83	60 - 83	60 - 83	1	Northern Star	24		A
	Duty Ball Charge	% v/v	13 - 17	10.5	10.5	10.5	10.5 - 11.3	1	Northern Star	16, 24		A
	Duty Total Load	% v/v	25 - 30	27	27	27	25	1	Northern Star	16, 24		A
	Total Ball Charge	Structural		18	18	18		1	Northern Star	23	For structural design	A
	Total Load	Structural		35	35	35		1	Northern Star	23	For structural design	A
	Mill Discharge Density	% w/w	75	75	75	75		1	Northern Star	9, 69		A
	Mill Pinion Power	Average	10,894	-	-	-		1	Northern Star	25		A
	Mill Pinion Power	Duty	11,682	16,757	15,078	16,757		6	Calculated			1
	Installed Motor Power		12,000	20,000	20,000	20,000	20,000	1	Northern Star	41, 23		A
	Liner Type			Steel	Steel	Steel	Steel	1	Northern Star	41		A
	Curved Pulp Lifter Capacity	Max		2,100	2,100	2,100	2,100	1	Northern Star	41		B
	Pebble Port Aperture		75	75	75	75		1	Northern Star	41	FLS offer	A
	Discharge Grate Open Area			11 - 12	11 - 12	11 - 12	11 - 12	1	Northern Star	41		B
	Trommel Dimensions	ID x L		3,829 m dia. X 3.80 m	3,829 m dia. X 3.80 m	3,829 m dia. X 3.80 m	3,829 m dia. X 3.80 m	5	Vendor	92		1
	Trommel Open Area	%	25	25	25	25		1	Northern Star	41		B
	Trommel Aperture			12 x 40	12 x 40	12 x 40	12 x 40	1	Northern Star	41		B
	Spray Water		150	150	150	150		2	Primero	40		B
	Trommel Undersize			1,050	1,050	1,050	1,050	2	Primero	32		B
	Existing Fim SAG Mill Transfer Density	% w/w	60					1	Northern Star	57, 69		B
4.3	Primary Mill Discharge Screen Discharge Screen Type			Horizontal Vibrating Screen	Horizontal Vibrating Screen	Horizontal Vibrating Screen	Horizontal Vibrating Screen	1	Northern Star	16		A
	Screen Size	W x L		4.27 x 8.54	4.27 x 8.54	4.27 x 8.54	4.27 x 8.54	5	Vendor	59		A
	Effective Screen Area	m ²		36.47	36.47	36.47	36.47	5	Vendor	59		B
	Screen Open Area	%		34.0	34.0	34.0	34.0	5	Vendor	59		B
	Screen Aperture	mm		12 x 55	12 x 55	12 x 55	12 x 55	2	Primero	40		A
	Specific Screen Flux Rate	m ³ /m ² /h OA		18.68	18.68	18.68	28.90	5	Vendor	59	Includes wash water	B
	Oversize Moisture	%w/w	5	13	13	13		5	Vendor	59		B
	Spray/Wash Water	m ³ /h		197.2	197.2	197.2	197.2	5	Vendor	59		B
	Discharge Screen Feed % Solids	%w/w		85	85	85		2	Primero	32		B
	Discharge Screen Oversize	t/h		375	375	375	375	1	Northern Star	41		B
4.4	Secondary Mill Type			Overflow Ball Mill	Overflow Ball Mill	Overflow Ball Mill	Overflow Ball Mill	1	Northern Star	41	NSR Supplied	A
	Mill Diameter			8.53	8.53	8.53	8.53	1	Northern Star	23	NSR Supplied	A
	Effective Grinding Length - EGL			13.26	13.26	13.26	13.26	1	Northern Star	23	NSR Supplied	A
	Duty Mill Speed	% Cs		60 - 78	60 - 78	60 - 78	60 - 78	1	Northern Star	41		A
	Duty Ball Charge	Nominal		28.5	28.5	28.5	28.5	1	Northern Star	41		A
	Duty Ball Charge	Maximum		31.0	31.0	31.0	31.0	1	Northern Star	24		A
	Total Ball Charge	Structural		38	38	38	38	1	Northern Star	23	With polymet liners. For structural design	A
	Mill Discharge Density	% w/w		75	75	75		1	Northern Star	4		A
	Mill Pinion Power	Duty		17,747	15,969	17,747	17,759	6	Calculated			
	Installed Motor Power			20,000	20,000	20,000	20,000	1	Northern Star	23		A
	Liner Type			Poly-Met	Poly-Met	Poly-Met	Poly-Met	1	Northern Star	23		A
	Trommel Flux Rate	Nominal		223	223	223	225	6	Calculated			
	Trommel Dimensions	ID x L		4,992 x 5.80	4,992 x 5.80	4,992 x 5.80	4,992 x 5.80	5	Vendor	92	ID = internal diameter	1
	Trommel Capacity	Maximum		6,681	6,681	6,681	6,681	1	Northern Star	23	Includes spray water	A
	Mill Discharge Flowrate	Nominal		6,373	6,197	6,383	6,277	6	Calculated		Includes spray water	
	Trommel Open Area	Minimum		30	30	30		1	Northern Star	23		A
	Trommel Aperture			19 x 40 - slotted	19 x 40 - slotted	19 x 40 - slotted	19 x 40 - slotted	1	Northern Star	23		A
	Trommel Screen Spray Water			150	150	150		1	Northern Star	4		A
4.5	Pebble Crusher Type		Cone	Cone	Cone	Cone	Cone	1	Northern Star	16	NSR Supplied	A
	Number		2 (duty / standby)	2 (duty / duty)	2 (duty / duty)	2 (duty / duty)	2 (duty / duty)	1	Northern Star	16, 23	NSR Supplied	A
	Model		Symons 78	XL7 SH-HD	XL7 SH-HD	XL7 SH-HD	XL7 SH-HD	1	Northern Star	41	NSR Supplied	A
	Installed Motor Power per Crusher		375 / 260	375	375	375	375	1	Northern Star	16, 41, 44		A
	Closed Side Setting	Minimum	7 - 9	7	7	7		1	Northern Star	16, 41		B
	Pebble Crushing Work Index	Maximum		23	23	23	23	1	Northern Star	56		1
	Pebble Bulk Density	Range		1.6 - 1.9	1.6 - 1.9	1.6 - 1.9	1.6 - 1.9	1	Northern Star	56		1
	Throughput	Nominal		302				6	Calculated			
	Throughput	Nominal		302				1	Northern Star			
		Nominal						1	Northern Star	41		B
		Range		23 - 27	305	305	610	1	Northern Star	16		B
	% of new feed							1	Northern Star	16		B
	% of new feed							1	Northern Star	16		B
	% of new feed							6	Calculated			
	Crusher Capacity at CSS	Maximum		610	610	610	610	1	Northern Star	41	305 t/h limit per crusher unit at CSS. This throughput represents the increased pebble load as the pebble ports wear out on SAG mill discharge grate	A
	Feed Bin Capacity	per bin		50	12.8	12.8	6	1, 2	Northern Star, Primero	26		A
		t		260	65	65	65	6	Calculated		Total Bin capacity of 130 t (2 bins)	
	Feed Moisture	Range		132	34	34	34	6	Calculated		Total Bin capacity of 68 m ³ (2 bins)	
	Feed Pebble PSD							1	Northern Star	41		B
	F ₁₀₀				95.0	95.0	95.0	1	Northern Star	41	Linear interpolation from PSD data	B
	F ₈₀				49.0	49.0	49.0	1	Northern Star	41	Linear interpolation from PSD data	B
	F ₅₀				37.0	37.0	37.0	1	Northern Star	41	Linear interpolation from PSD data	B
	Product Pebble PSD											
	P ₈₀			12.0	12.0	12.0	12.0	1	Northern Star	69	FLS guaranteed value	B
4.6	Classification Feed Size F ₅₀	Typical		276	276	276	276	1	Northern Star	16		A
	Feed Size F ₁₀₀	Modelled		350	350	350	350	1	Northern Star	9	OMC Modelling	B
	Feed Size F ₈₀	Modelled		0.757	0.757	0.757	0.757	1	Northern Star	9	OMC Modelling	B
	Feed Size F ₅₀	Modelled		0.306	0.306	0.306	0.306	1	Northern Star	9	OMC Modelling	B
	Product Size P ₈₀	Target		180	180	180	220	1	Northern Star	26		A
	Product Size P ₈₀	Maximum		250	250	250	250	1	Northern Star	26		A
	Type			Hydrocyclone	Hydrocyclone	Hydrocyclone	Hydrocyclone	1	Northern Star	16		A
	Cyclone Model			800CVX13	800CVX13	800CVX13	800CVX13	5	Vendor	28		A
	Number of Cyclone Clusters			1	1	1	1	5	Vendor	28		A
	Distributor Outlets per Cluster	No.		18	18	18	18	5	Vendor	28		B
	Number Installed Cyclones			17	17	17	17	5	Vendor	28	15 for primary cyclone duty, 2 for gravity duty	A
	Number Spare Outlets			1	1	1	1	6	Calculated		1 blanked nozzle for the ladder access, 1 spare blanked nozzle for future	
	Number Operating Cyclones - Primary Cluster	Nominal		12	11							

AREA	PARAMETER		UNIT	DATA				REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV	
				1	2	3	4						
				FIMISTON	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um						
4.7	Number Operating Cyclones - Gravity	Nominal	No.		2	2	2	5	Vendor	28	Gravity cyclones require larger spigot and VF to achieve mass flow of 1000tph to underflow. There is also a higher recirc. load.	A	
		Maximum	No.		2	2	2	5	Vendor	28		A	
	Cyclone Feed	Nominal	th		14,421	13,986	14,421	6	Calculated				
	Cyclone Feed Flow	Nominal	m³/h		13,233	12,816	13,217	6	Calculated				
	Cyclone Feed Flow	Maximum @ 61.6%	m³/h		13,862	13,862	13,862	1	Northern Star	26	Flow rate of 13,862m³/hr requires 14 or 15 online. 2 gravity as advised by Weir	A	
	Cyclone Feed Density		% w/w		62.0	62.0	62.0	5	Vendor	28		A	
	Underflow Solids Concentration	Minimum	% w/w		75.0	75.0	75.0	1	Northern Star	16		B	
	Underflow Solids Concentration		% w/w		77.2	77.0	77.2	6	Calculated				
	Underflow Solids		th		11,275	11,024	11,275	6	Calculated				
	Overflow Solids Concentration		% w/w		36.4	35.9	36.4	5	Vendor	28		A	
	Overflow Solids		th		3,146	2,962	3,146	6	Calculated				
	Circulating Load - Cyclone UF / Cyclone OF		%		358	372	358	5	Vendor	28		A	
	Circulating Load - Ball Mill Discharge / Cyclone OF		%		301	311	301	6	Calculated				
	Operating Pressure - Cyclones in Cluster		kPa		82	91	82	5	Vendor	28		B	
	Operating Pressure - Gravity Cyclones		kPa		83	86	83	5	Vendor	28		B	
	Trash Screen												
	Type				Linear Screen	Linear Screen	Linear Screen	Linear Screen	1	Northern Star	16		A
	Total Number Screens				4	4	4	4	6	Calculated			
	Number Operating				3	3	3	3	1	Northern Star	14		A
	Number Standby				1	1	1	1	1	Northern Star	14		A
	Screen Dimensions	W x L	m		9.53 x 4.2	9.53 x 4.2	9.53 x 4.2	9.53 x 4.2	5	Vendor	55		B
	Screen Area per Screen	Effective	m²		40.03	40.03	40.03	40.03	5	Vendor	55		A
	Total Screen Area		m²		120.09	120.09	120.09	120.09	6	Calculated			
	Effective Screen Area		%		34	34	34	34	5	Vendor	55		B
	Screen Aperture		mm		1.0 x 1.0	1.0 x 1.0	1.0 x 1.0	1.0 x 1.0	5	Vendor	55		B
	Screen Feed Flow (incl. spray water)		m³/h		2,183	2,093	2,182	1,936	6	Calculated			
	Maximum Screen Flux Rate		m³/m²/h		75	75	75	75	1	Northern Star	26		A
	Specific Screen Flux Rate		m³/m²/h		54.5	52.3	54.5	48.4	6	Calculated			
	Spray/Wash Water	per screen	m³/h		60	60	60	60	5	Vendor	31, 55		B
	Total Spray/Wash Water		m³/h		180	180	180	180	6	Calculated			
	Spray Water reporting to Screen Oversize	per screen	m³/h		45	45	45	45	5	Vendor	31		B
	Total Spray Water reporting to Screen Oversize	Total	m³/h		135	135	135	135	6	Calculated			
	Spray Water reporting to Screen Undersize	per screen	m³/h		15	15	15	15	5	Vendor	31		B
	Spray Water reporting to Screen Undersize	Total	m³/h		45	45	45	45	6	Calculated			
	Water to Trash Screens Final Collection Box		m³/h		258.0	134.9	258.0	306.8	6	Calculated		HP Process Water, back calculated in SysCAD	B
Trash Dewatering													
Dewatering Screen Type				Vibrating Screen	Vibrating Screen	Vibrating Screen	Vibrating Screen	1	Northern Star	16		A	
Number				1	1	1	1	1	Northern Star	16		A	
Screen Size	W x L	m		1.83 x 3.66	1.83 x 3.66	1.83 x 3.66	1.83 x 3.66	5	Vendor	59		B	
Effective Screening Area per Screen		m²		6.14	6.14	6.14	6.14	5	Vendor	59		B	
Specific Screen Flux Rate	Design	m³/m²/h		50	50	50	50	1	Northern Star	16		A	
	Nominal	m³/m²/h		22.36	22.36	22.36	22.36	5	Vendor	59	Includes wash water.	A	
Screen Aperture		mm		1 x 15	1 x 15	1 x 15	1 x 15	1	Northern Star	16		A	
Deck Type				Polyurethane	Polyurethane	Polyurethane	Polyurethane	5	Vendor	59		A	
Quantity of Wood/Grit/Trash removed per day	td/day			27	27	27	27	1	Northern Star	16		A	
Total Spray/Wash Water		m³/h		11	11	11	11	5	Vendor	59		B	
5.0 GRAVITY CONCENTRATION													
5.1	Gravity Scalping Screen												
	Type			Horizontal Vibrating	Horizontal Vibrating	Horizontal Vibrating	Horizontal Vibrating	5	Vendor	59		A	
	Number of Screens			2	2	2	2	1	Northern Star	16	Only 2x gravity concentrator circuits, no future allowance for additional circuits	A	
	Screen Solids Feed Rate per Screen	Nominal	th	1,000	1,000	1,000	1,000	6	Calculated			A	
		Design	th	1,111	1,111	1,111	1,111	6	Calculated			A	
	Screen Feed Density	Target	% w/w	60.0	60.0	60.0	60.0	1	Northern Star	70	Excludes Spray Water	B	
	Screen Feed Density	Actual	% w/w	60.0	60.0	60.0	60.0	6	Calculated				
	Screen Feed Flowrate per Screen	Nominal	m³/h	968	966	967	967	6	Calculated		Excludes Spray Water		
		Design	m³/h	1,075	1,073	1,074	1,075	6	Calculated		Excludes Spray Water		
	Screen Size	W x L	m	3.66 x 6.10	3.66 x 6.10	3.66 x 6.10	3.66 x 6.10	1	Northern Star	59, 70, 72		B	
	Effective Screen Area per Screen	Overall	m²	22.31	22.31	22.31	22.31	5	Vendor	59			
	Specific Screen Flux Rate	Nominal	m³/m²/h	48.0	47.9	47.9	47.9	6	Calculated		Includes Spray Water		
		Design	m³/m²/h	52.8	52.7	52.7	52.7	6	Calculated		Includes Spray Water		
		Target	m³/m²/h	50	50	50	50	1	Northern Star	16, 70		A	
	Screen Split to Oversize - Range		%	8 - 15	8 - 15	8 - 15	8 - 15	1	Northern Star	16		A	
		Design	%	10	10	10	10	1	Northern Star	16, 72		A	
	Screen Aperture		mm	3.2 x 18mm SWF first half & 2.0 x 18mm SWF second half	3.2 x 18mm SWF first half & 2.0 x 18mm SWF second half	3.2 x 18mm SWF first half & 2.0 x 18mm SWF second half	3.2 x 18mm SWF first half & 2.0 x 18mm SWF second half	5	Vendor	59	3.2mm Slotted first half, 2mm non-pegging second half (VR) requested by NSR in source 16	B	
	Screen Spray Water	per screen	m³/h	102	102	102	102	5	Vendor	59		B	
	Oversize Solids Concentration		% w/w	87	87	87	87	5	Vendor	59		B	
	Oversize Slurry Destination			Ball mill feed	Ball mill feed	Ball mill feed	Ball mill feed	1	Northern Star	43		A	
	Undersize Slurry Destination			Gravity Concentrators	Gravity Concentrators	Gravity Concentrators	Gravity Concentrators	1	Northern Star	43		A	
5.2	Gravity Concentrator												
	Type			Knelson	Knelson	Knelson	Knelson	1	Northern Star	16		A	
	Number of Concentrators			2	2	2	2	1	Northern Star	16	Only 2x gravity concentrator circuits	A	
	Model			KC-QS70	KC-QS70	KC-QS70	KC-QS70	1	Northern Star	16		A	
	Feed Capacity per Unit	Max	th	1,000	1,000	1,000	1,000	5	Vendor	39, 71			A
	Volumetric Capacity per Unit	Max	m³/h	910	910	910	910	5	Vendor	7		B	
	Feed Solids Rate to Circuit		th	1,800	1,800	1,800	1,800	2, 5	Primero, Vendor	7, 71	300-1000t/hr/unit operating range	A	
	Feed Solids Rate per Unit	Max	th	2,000	2,000	2,000	2,000	6	Calculated				
	Feed Solids Rate per Unit	Nominal	th	900	900	900	900	6	Calculated		Vendor recommends 90% of the max (Source 71).		
	Feed Density		% w/w	54.2	54.2	54.2	54.2	6	Calculated				
	Volumetric Flowrate per Unit		m³/h	1,021	1,020	1,020	1,021	6	Calculated				
	Fluidising Water per Unit		m³/h	120	120	120	120	5	Vendor	7		B	
	Fluidising Water	Nominal	m³/h	240	240	240	240	6	Calculated		120m3 per concentrator		
	Cycle Time (Inclusive of Flush Cycle)		min	30	30	30	30	5	Vendor	71		B	
	Cycles Per Day		No.	48	48	48	48	6	Calculated				
	Concentrate Production Per Cycle		kg	150	150	150	150	5	Vendor	7, 71		A	
	Concentrate Bulk SG		tim³	2.20	2.20	2.20	2.20	1	Northern Star	44		B	
	Daily Concentrate Production		kg/d	14,400	14,400	14,400	14,400	6	Calculated				
			m³/d	6.55	6.55	6.55	6.55	6	Calculated				
	Fresh Feed to Concentrators		%	57.2	60.8	57.2	51.5	6	Calculated				
	Tailings Destination			Cyclone Feed Hopper	Cyclone Feed Hopper	Cyclone Feed Hopper	Cyclone Feed Hopper	1	Northern Star	43		A	
6.0 INTENSIVE LEACH													
6.1	Intensive Leach Reactor												
	Type			Acacia	Acacia	Acacia	Acacia	1	Northern Star	16		A	
	Model			CS10000	CS10000	CS10000	CS10000	1	Northern Star	70		B	
	Number			1	1	1	1	1	Northern Star	16		A	
	Batches per Day	Nominal		1	1	1	1	5	Vendor	39		A	
		Max		2	2	2	2	5	Vendor	39	Required to complete 2 batches per day as per NSR Source 72	A	
	Solids Feed Rate per Batch		kg/batch	14,400	14,400	14,400	14,400	6>					

AREA	PARAMETER	UNIT	DATA				REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV
			1	2	3	4					
			FIMISTON	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um					
7.0	FLOTATION										
7.1	General										
	Throughput	t/h		3,146	2,962	3,146	3,494	6	Calculated		
	Feed Particle Size - p80	µm		180	180	180	220	1	Northern Star		A
		Design		220	220	220	220	1	Northern Star	26	B
	Froth Factors										
	Rougher Flotation Concentrate			2.00	2.00	2.00	2.00	1	Northern Star	16	A
	Scavenger Flotation Concentrate			1.60	1.60	1.60	1.60	1	Northern Star	16	A
	Cleaner Flotation Concentrate			2.50	2.50	2.50	2.50	1	Northern Star	16	A
	Cleaner Scavenger Flotation Concentrate			2.00	2.00	2.00	2.00	1	Northern Star	16	A
	Flotation Cell Transfer Lines			1.20	1.20	1.20	1.20	1	Northern Star	16	A
	Launders			-	-	-	-	1	Northern Star	16	A
	Spray Water Line Pressure (at vendor battery limit)	kPa		62.74	62.74	62.74	62.74	5	Vendor	60	B
7.2	Rougher Flotation										
	Feed Streams			Trash Screen Underflow	Trash Screen Underflow	Trash Screen Underflow	Trash Screen Underflow	1	Northern Star	43	
	Solids Feed Rate	New Feed	t/h	3,146	2,962	3,146	3,494	6	Calculated		A
		Total	t/h	3,161	2,977	3,160	3,512	6	Calculated		
	Volumetric Feed Rate		m³/h	6,719	6,326	6,716	6,027	6	Calculated		
	Feed Solids Concentration		% w/w	34.4	34.4	34.4	40.5	6	Calculated		
	Particle Size p80	Design	µm	220	220	220	220	2	Primero	63	B
	Mass Pull		%	4.4	4.4	4.4	4.4	6	Calculated		
	Concentrate 1 Upgrade Ratio		% S	15	15	15	15	1	Northern Star	16	A
	Concentrate 1 Grade		% Au in fl. feed	15.3	22.4	21.0	15.3	6	Calculated		
	Concentrate 1 Stage Recovery		% of S in fl. feed	65	65	65	65	1	Northern Star	23	A
	Concentrate 1 Production		t/h	65	65	65	65	1	Northern Star	23	A
	Concentrate 1 Solids Concentration		% w/w	138	130	138	153	6	Calculated		
	Concentrate 1 Slurry Volume - Excl Spray Water		m³/h	25	25	25	25	1	Northern Star	16	A
	Concentrate 1 SG			427	400	425	474	6	Calculated		
	Concentrate 1 Launder Spray Water	Operating Range	m³/h	3.32	3.54	3.50	3.32	2	Primero	32	A
		Design	m³/h	6.98 - 13.00	6.98 - 13.00	6.98 - 13.00	6.98 - 13.00	5	Vendor	60	B
				50	50	50	50	5	Vendor	60	A
	Mass Pull		%	2.4	2.4	2.4	2.4	6	Calculated		
	Concentrate 2 & 3 Upgrade Ratio		% S	7	7	7	7	1	Northern Star	16	A
	Concentrate 2 & 3 Grade		% Au in fl. feed	7.1	10.4	9.8	7.1	6	Calculated		
	Concentrate 2 & 3 Stage Recovery		% of S in fl. feed	17	17	17	17	1	Northern Star	23	A
	Concentrate 2 & 3 Production		t/h	17	17	17	17	1	Northern Star	23	A
	Concentrate 2 & 3 Solids Concentration		% w/w	77	72	77	86	6	Calculated		
	Concentrate 2 & 3 Slurry Volume - Excl Spray Water		m³/h	25	25	25	25	1	Northern Star	16	A
	Concentrate 2 & 3 SG			241	225	239	268	6	Calculated		
	Concentrate 2 Launder Spray Water	Operating Range	m³/h	3.06	3.17	3.15	3.06	2	Primero	32	A
	Concentrate 3 Launder Spray Water	Operating Range	m³/h	6.98 - 13.00	6.98 - 13.00	6.98 - 13.00	6.98 - 13.00	5	Vendor	60	
	Concentrate 2 & 3 Spray Water	Design (Combined)	m³/h	6.98 - 13.00	6.98 - 13.00	6.98 - 13.00	6.98 - 13.00	5	Vendor	60	A
				50	50	50	50	5	Vendor	50	
	Rougher Tail Solids Rate		t/h	2,946	2,775	2,946	3,273	6	Calculated		
	Rougher Tail Slurry Volumetric Flow		m³/h	6,051	5,701	6,052	5,285	6	Calculated		
	Cell Type			Tank Cells	Tank Cells	Tank Cells	Tank Cells	1	Northern Star	16	A
	Aspiration			Forced	Forced	Forced	Forced	1	Northern Star	16	A
	Number Cells Installed			3	3	3	3	1	Northern Star	41	A
	Nominal Cell Volume		m³	630	630	630	630	1	Northern Star	16	A
	Total Cell Volume		m³	1,890	1,890	1,890	1,890	6	Calculated		
	Cell Air Hold-Up		%	15	15	15	15	1	Northern Star	41	A
	Plant Residence Time		min	15.1	16.0	15.1	17.0	6	Calculated		
	Lip Loading - Design		t/m/h	1.5	1.5	1.5	1.5	1	Northern Star	16	A
	Froth Carry Rate - Design		t/m²/h	1.5	1.5	1.5	1.5	1	Northern Star	16	A
7.3	Scavenger Flotation										
	Feed Streams			Rougher Tailings	Rougher Tailings	Rougher Tailings	Rougher Tailings	1	Northern Star	43	
	Solids Feed Rate		t/h	2,946	2,775	2,946	3,273	6	Calculated		A
	Volumetric Feed Rate		m³/h	6,051	5,701	6,052	5,286	6	Calculated		
	Feed Solids Concentration		% w/w	35.3	35.3	35.3	42.4	6	Calculated		
	Particle Size p80	Design	µm	220	220	220	220	2	Primero	63	B
	Mass Pull (Scavenger Concentrate)		%	1.1	1.1	1.1	1.1	6	Calculated		
	Scavenger Concentrate Upgrade Ratio		% S	5	5	5	5	1	Northern Star	16	A
	Scavenger Concentrate Grade		% Au in fl. feed	5.1	7.5	7.0	5.1	6	Calculated		
	Scavenger Concentrate Stage Recovery		% of S in fl. feed	5	5	5	5	1	Northern Star	23	A
	Scavenger Concentrate Stage Recovery		% of S in fl. feed	5	5	5	5	1	Northern Star	23	A
	Scavenger Concentrate Production		t/h	32	30	32	35	6	Calculated		
	Scavenger Concentrate Solids Concentration		% w/w	25	25	25	25	2	Primero	32	A
	Scavenger Concentrate Slurry Volume - Excl Spray Water		m³/h	99	93	99	110	2	Calculated		
	Scavenger Concentrate SG			3.00	3.07	3.06	3.00	2	Primero	32	A
	Scavenger Concentrate Solids Concentration - Diluted		% w/w	20	20	20	20	2	Primero	32	A
	Scavenger Concentrate Launder Spray Water	Actual (Combined)	m³/h	29	28	29	33	6	Calculated		
		Operating Range	m³/h/cell	6.98 - 13.00	6.98 - 13.00	6.98 - 13.00	6.98 - 13.00	5	Vendor	60	B
		Design	m³/h/cell	25.00	25.00	25.00	25.00	5	Vendor	60	B
	Scavenger Tail Solids Rate		t/h	2,914	2,745	2,914	3,238	6	Calculated		
	Scavenger Tail Slurry Volumetric Flow		m³/h	5,952	5,608	5,953	5,176	6	Calculated		
	Cell Type			Tank Cells	Tank Cells	Tank Cells	Tank Cells	1	Northern Star	16	A
	Aspiration			Forced	Forced	Forced	Forced	1	Northern Star	16	A
	Number Cells Installed			4	4	4	4	1	Northern Star	41	A
	Nominal Cell Volume		m³	630	630	630	630	1	Northern Star	16	A
	Total Cell Volume		m³	2,520	2,520	2,520	2,520	6	Calculated		
	Cell Air Hold-Up		%	15	15	15	15	1	Northern Star	41	A
	Plant Residence Time		min	21.4	22.7	21.4	24.6	6	Calculated		
	Lip Loading - Design		t/m/h	1.5	1.5	1.5	1.5	1	Northern Star	16	A
	Froth Carry Rate - Design		t/m²/h	1.5	1.5	1.5	1.5	1	Northern Star	16	A
	Combined Rougher and Scavenger Residence Time			36.5	38.8	36.5	41.6	6	Calculated		
7.4	Cleaner Flotation										
	Feed Streams			Rougher and Scavenger Concentrate Cleaner Scavenger Concentrate	Rougher and Scavenger Concentrate Cleaner Scavenger Concentrate	Rougher and Scavenger Concentrate Cleaner Scavenger Concentrate	Rougher and Scavenger Concentrate Cleaner Scavenger Concentrate	1	Northern Star	43	
	Solids Feed Rate	New Feed Only	t/h	253	244	258	282	6	Calculated		A
		Total	t/h	465	424	439	478	6	Calculated		
	Volumetric Feed Rate to Cell - Target		m³/h	1,998	1,998	1,998	1,998	5	Vendor	27	C
	Volumetric Feed Rate to Cell - Actual		m³/h	1,998	1,998	1,998	1,998	6	Calculated		
	Feed Solids Concentration - Actual		% w/w	19.0	17.5	18.0	19.5	6	Calculated		
	Particle Size p80	Design	µm	220	220	220	220	6	Calculated		
		Calculated	µm	180	180	180	220	6	Calculated		
	Mass Pull		%	30.8	44.0	41.5	30.7	6	Calculated		
	Concentrate Grade		% S	35	35	35	35	1	Northern Star	26	A
	Concentrate Stage Recovery		% S	93.3	93.6	93.4	93.1	6	Calculated		
	Concentrate Recovery	Overall	% of S in fl. feed	85.1	85.1	85.1	85.1	1	Northern Star	23	A
	Concentrate Recovery	Overall	% Au in fl. feed	84	83	83	84	6	Calculated		
	Concentrate Production		t/h	78	107	107	86	6	Calculated		
	Concentrate Solids Concentration		% w/w	25	25	25	25	5	Vendor	27	A
	Concentrate Slurry Volume - Excl Spray Water		m³/h	237	326	326	263	6	Calculated		
	Concentrate SG			3.93	3.93	3.93	3.93	2	Primero	32	A
	Concentrate Wash Water Ratio (of Con m3/hr)			1.0	1.0	1.0	1.0	5	Vendor	27	A
	Cleaner Tail Solids Rate		t/h	183	145	159	203	6	Calculated		
	Cleaner Tail Slurry Volumetric Flow		m³/h	945	910	957	1,038	6	Calculated		
	Cleaner Tails Recycle		%	52.7	54.4	52.1	48.1	6	Calculated		
	Cleaner Tail Solids Rate		t/h	204	173	173	188	6	Calculated		
	Cleaner Tail Slurry Volumetric Flow		m³/h	1,053	1,068	1,041	960	6	Calculated		
	Cell Type			Jameson	Jameson	Jameson	Jameson	5	Vendor	18	A
	Model			B6500/24	B6500/24	B6500/24	B6500/24	5	Vendor	27	A
	Aspiration Type			Natural	Natural	Natural	Natural	5	Vendor	16	A
	Number Cells Installed			1	1	1	1	5	Vendor	18	A
	Number of Downcomers			24	24	24	24	5	Vendor	27	B
	Froth Area		m²	22.2	22.2	22.2	22.2	5	Vendor	27	A
	Froth Carry Rate	Nom	t/m²/h	3.50	4.81	3.50	3.51	5	Vendor	27	B
		Max	t/m²/h	6.5	6.5	6.5	6.5	5	Vendor	27	A
		Calculated	t/m²/h	3.51	4.82	4.82	3.89	6	Calculated		

AREA	PARAMETER	UNIT	DATA				REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV	
				1	2	3	4					
			FIMISTON	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um	NEW CIRCUIT 28.35 MTPA THROUGHPUT NOMINAL GRADE P80 220um					
7.5	Cleaner Scavenger Flotation											
	Feed Streams	New Feed Only	th	Cleaner Flotation Tail	Cleaner Flotation Tail	Cleaner Flotation Tail	Cleaner Flotation Tail	1	Northern Star	66	NSR Supplied	B
	Solids Feed Rate		th	183	145	159	203	6	Calculated		New feed only	
		Total	th	380	300	317	385				Total feed to Jameson cell including tails recycle	
	Volumetric Feed Rate to Cell - Target		m³/h	2,048	2,048	2,048	2,048	5	Vendor	27	Average of four modelled cases	C
	Volumetric Feed Rate - Actual		m³/h	2,048	2,048	2,048	2,048	6	Calculated		Total feed to Jameson cell including tails recycle	
	Feed Solids Concentration	Total	% w/w	15.6	12.6	13.2	15.8	6	Calculated		Total feed to Jameson cell including tails recycle	
	Particle Size p80	Design	µm	220	220	220	220	2	Primero	63	Feed, concentrate and tailings p80 assumed equal (conservative). Max of calculated p80	
	Mass Pull		%	7.6	13.3	12.1	7.6	6	Calculated			
	Concentrate Grade		% S	13.10	13.10	13.10	13.10	5	Vendor	19	Design Grade sulphur grade based on Primero circuit modelling	A
	Concentrate Stage Recovery		% S	49.6	54.1	54.2	50.9	6	Calculated			A
	Concentrate Recovery	% of Au in fl. feed		5.7	5.7	5.7	5.7	3	Assumed		Assumed to match the S recovery as seen in other areas of the flotation circuit	A
	Concentrate Recovery	% of S in fl. feed		5.7	5.7	5.7	5.7	2	Primero	32		A
	Concentrate Production		th	14	19	19	15	6	Calculated			
	Concentrate Solids Concentration		% w/w	25	25	25	25	2	Primero	32		B
	Concentrate Slurry Volume - Excl Spray Water		m³/h	43	60	60	48	6	Calculated			
	Concentrate SG			3.25	3.25	3.25	3.25	2	Primero	32		A
	Concentrate Wash Water Ratio (of Con m3/hr)			1	1	1	1	5	Vendor	27		A
	Cleaner Scavenger Tail Solids Rate		th	169	125	140	188	6	Calculated		To Cleaner-Scavenger Flotation Cell Tails Hopper	
	Cleaner Scavenger Tail Slurry Volumetric Flow		m³/h	948	914	960	1,041	6	Calculated		To Cleaner-Scavenger Flotation Cell Tails Hopper	
	Cleaner Scavenger Tails Recycle		%	53.7	55.4	53.1	49.2	6	Calculated		The % of total tailings recycled back to feed hopper.	
	Cleaner Scavenger Tail Solids Rate		th	196	158	168	192	6	Calculated		Recycle back to Cleaner Flotation Cell Feed Hopper	
	Cleaner Scavenger Tail Slurry Volumetric Flow		m³/h	1,100	1,134	1,088	1,007	6	Calculated		Recycle back to Cleaner Flotation Cell Feed Hopper	
	Cell Type			Jameson	Jameson	Jameson	Jameson	5	Vendor	19		A
	Model			B6500/24	B6500/24	B6500/24	B6500/24	1	Northern Star	27		A
	Aspiration Type			Natural	Natural	Natural	Natural	1	Northern Star	16		A
	Number Cells Installed			1	1	1	1	5	Vendor	19		A
	Number of Downcomers			24	24	24	24	5	Vendor	27		B
	Froth Area		m²	22.2	22.2	22.2	22.2	5	Vendor	27		A
	Froth Carry Rate	Nom	l/m²/h	0.63	0.85	0.63	0.63	5	Vendor	27		B
		Max	l/m²/h	6.5	6.5	6.5	6.5	5	Vendor	27		A
		Calculated	l/m²/h	0.63	0.86	0.86	0.70	6	Calculated			
8.0	CONCENTRATE HANDLING											
8.1	Concentrate Thickener (Brownfields)			Existing Flotation Tail Pre-Leach Thickener (CIL3)	Existing Flotation Tail Pre-Leach Thickener (CIL3)	Existing Flotation Tail Pre-Leach Thickener (CIL3)	Existing Flotation Tail Pre-Leach Thickener (CIL3)				Existing 30-TH-61	
	Feed Stream			Final Concentrate Hopper (Cleaner Flotation Concentrate)	Final Concentrate Hopper (Cleaner Flotation Concentrate)	Final Concentrate Hopper (Cleaner Flotation Concentrate)	Final Concentrate Hopper (Cleaner Flotation Concentrate)	1	Northern Star	16		A
	Solids Feed Rate		th	75	104	104	84	6	Calculated			
	Feed Flowrate - Excl. Diluted Flocculant		m³/h	230	319	319	256	6	Calculated			
	Feed Solids Concentration		% w/w	24.9	24.9	24.9	24.9	6	Calculated			
	Underflow Solids Concentration		% w/w	60	60	60	60	1	Northern Star	16		A
	Testwork Flux Rate		l/m²/h	0.25	0.25	0.25	0.25	1	Northern Star	23		A
	Calculated Flux Rate		l/m²/h	0.13	0.18	0.18	0.15	6	Calculated			
	Thickener Area Calculated		m²	573	573	573	573	6	Calculated			
	Diameter	Existing	m	27	27	27	27	1	Northern Star	16		A
	Structural Design SG			3.8	3.8	3.8	3.8	1	Northern Star	16		A
	Concentrate Thickener Sprays - Mill Circuit Water		m³/h	68.40	68.40	68.40	68.40	5	Vendor	113		1
8.2	Filtered Feed Tanks A & B (Brownfields)			Existing	Existing	Existing	Existing					
	Feed Streams			Concentrate Thickener Underflow	Concentrate Thickener Underflow	Concentrate Thickener Underflow	Concentrate Thickener Underflow	1	Northern Star	43	Shared volume between Filtered Feed Tank A & B via balance line.	A
	Solids Feed Rate		th	75	104	104	84	6	Calculated			
	Volumetric Feed Rate		m³/h	66	92	92	74	6	Calculated			
	Feed Solids Concentration		% w/w	59.6	59.7	59.7	59.6	6	Calculated			
	Number of Tanks			2	2	2	2	1	Northern Star	16		A
	Residence Time		h	21.2	15.3	15.3	19.0	6	Calculated			
	Tank Capacity	per tank	m³	700	700	700	700	1	Northern Star	16		A
8.3	Concentrate Filters (Brownfields)			Existing	Existing	Existing	Existing				Remains in operation to treat additional concentrate in excess of the UFG capacity.	
	Feed Stream			Concentrate Filtered Feed Tank B	Concentrate Filtered Feed Tank B	Concentrate Filtered Feed Tank B	Concentrate Filtered Feed Tank B	1	Northern Star	43		A
	Filter Type			Vacuum Belt Filter	Vacuum Belt Filter	Vacuum Belt Filter	Vacuum Belt Filter	1	Northern Star	43		A
	Number of Filters			2	2	2	2	1	Northern Star	16		A
	Solids Feed Rate		th	0	0	0	0	6	Calculated		Solids have been prioritised through UFG circuits first.	
	Volumetric Feed Rate		m³/h	0	0	0	0	6	Calculated			
	Feed Solids Concentration	Target	% w/w	60.0	60.0	60.0	60.0	1	Northern Star	16		A
		Actual	% w/w	0.0	0.0	0.0	0.0	6	Calculated			
	Target Moisture Content		<% w/w	10.0	10.0	10.0	10.0	1	Northern Star	16		A
	Filter Area	per filter	m²	55.0	55.0	55.0	55.0	1	Northern Star	16		A
	Specific Filtration Rate	Nominal	m³/m²/h	0.14	0.14	0.14	0.14	1	Northern Star	16		A
		Design	m³/m²/h	0.31	0.31	0.31	0.31	1	Northern Star	16		A
8.4	Concentrate Stockpile			Concentrate Filter Cake	Concentrate Filter Cake	Concentrate Filter Cake	Concentrate Filter Cake				Remains in operation to treat additional concentrate in excess of the UFG capacity.	
	Feed Stream			1	1	1	1	1	Northern Star	43		A
	No. of Stockpiles			1	1	1	1	1	Northern Star	16		A
	Concentrate Bulk Density		l/m³	2.2	2.2	2.2	2.2	1	Northern Star	16		A
	Angle of Repose	Nominal	Degrees	40.0	40.0	40.0	40.0	1	Northern Star	16		A
	Storage		days	2	2	2	2	1	Northern Star	16		A
	Capacity		m³	1,139	1,139	1,139	1,139	1	Northern Star	16		A
	TML for Bulk Shipment		% w/w	6.10 - 8.26	6.10 - 8.26	6.10 - 8.26	6.10 - 8.26	1	Northern Star	23		A
9.0	CONCENTRATE ULTRAFINE GRINDING											
9.1	Concentrate UFG Mill 1 (Brownfields)			Existing	Existing	Existing	Existing				M3000	
	Circuit Feed Streams			Concentrate Filter Feed Tank A	Concentrate Filter Feed Tank A	Concentrate Filter Feed Tank A	Concentrate Filter Feed Tank A	1	Northern Star	16, 80		1
	Feed Rate - Fresh Feed	P80 9 µm	th	0.0	10.0	10.0	0.0	1	Northern Star	69		B
		Design	th	0.0	13.0	13.0	0.0	1	Northern Star	16		A
	Feed Solids Concentration		% w/w	45	45	45	45	1	Northern Star	4		A
	Feed F80	Nominal	µm	144	144	144	144	1	Northern Star	16		A
	Specific Energy	P80 9 µm	kWh/t	116.5	116.5	116.5	116.5	1	Northern Star	16		A
	Type			IsaMill	IsaMill	IsaMill	IsaMill	1	Northern Star	16		A
	Model			M3000	M3000	M3000	M3000	1	Northern Star	16		A
	Number			1	1	1	1	1	Northern Star	16		A
	Configuration			Closed Circuit	Closed Circuit	Closed Circuit	Closed Circuit	1	Northern Star	16		A
	Mill Power Draw	Nominal	kW	860	860	860	860	1	Northern Star	16		A
	Installed Motor Power		kW	1,120	1,120	1,120	1,120	1	Northern Star	16		A
	UFG Mill Recycle Proportion		%	0	0	0	0	1	Northern Star	4		A
	UFG Circuit Product Size P80	Nominal	µm	9.0	9.0	9.0	9.0	1	Northern Star	16		A
	UFG Circuit Product Size P80 - Range		µm	9 - 12	9 - 12	9 - 12	9 - 12	1	Northern Star	16		A
	Gland Seal Water		m³/h	0.0	1.2	1.2	0.0	1	Northern Star	2		A
	Flushing Gland Seal Water		m³/h	0.0	5.4	5.4	0.0	1	Northern Star	2		A
9.2	UFG Mill 1 Trash Screen (Brownfields)			Existing	Existing	Existing	Existing					
	Feed Stream			UFG Mill 1 Discharge Vibrating Screen	UFG Mill 1 Discharge Vibrating Screen	UFG Mill 1 Discharge Vibrating Screen	UFG Mill 1 Discharge Vibrating Screen	1				

AREA	PARAMETER	UNIT	DATA				REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV
			1	2	3	4					
			FIMISTON	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um	NEW CIRCUIT 28.35 MTPA THROUGHPUT NOMINAL GRADE P80 220um				
9.4	New Concentrate UFG Mill 2 & 3			Greenfields Concentrate Filter Feed Tank A & B (common balance line)	Greenfields Concentrate Filter Feed Tank A & B (common balance line)	Greenfields Concentrate Filter Feed Tank A & B (common balance line)	Greenfields Concentrate Filter Feed Tank A & B (common balance line)			M15,000, 2 off	A
	Circuit Feed Streams	UFG Mill 2/3						2	Primerio		0
	Nominal Feed Rate - Fresh Feed	P ₈₀ 9 µm	t/h	30.0	30.0	30.0	30.0	1	Northern Star	24	A
	Maximum Feed Rate - Fresh Feed	P ₈₀ 12 µm	t/h	46.0	46.0	46.0	46.0	1	Northern Star	24	A
	Total Feed Solids Concentration		% w/w	47.5	47.5	47.5	47.7	5	Vendor	65	B
	Feed F ₈₀	Nominal	µm	144	144	144	144	1	Northern Star	16	A
	Specific Energy	P ₈₀ 9 µm	kWh/t	116.5	116.5	116.5	116.5	1	Northern Star	16	A
	Type			IsaMill	IsaMill	IsaMill	IsaMill	1	Northern Star	23	A
	Model			M15,000	M15,000	M15,000	M15,000	1	Northern Star	23	A
	Number			2	2	2	2	1	Northern Star	23, 43	A
	Configuration			Closed Circuit	Closed Circuit	Closed Circuit	Closed Circuit	1	Northern Star	23	A
	Mill Power Draw	Nominal	kW	3,500	3,500	3,500	3,500	1	Northern Star	5	A
	Installed Motor Power		kW	3,800	3,800	3,800	3,800	1	Northern Star	23	A
	UFG Mill Recycle Proportion		%	20	20	20	5	5	Vendor	65	B
	UFG Circuit Product Size P ₈₀	Nominal	µm	9.0	9.0	9.0	9.0	1	Northern Star	16	A
	UFG Circuit Product Size P ₈₀ - Range		µm	9 - 12	9 - 12	9 - 12	9 - 12	1	Northern Star	16	A
	Gland Seal Water		m³/h	2.0	2.0	2.0	2.0	5	Vendor	65	B
	Flushing Gland Seal Water		m³/h	5.4	5.4	5.4	5.4	5	Vendor	65	B
9.5	UFG Mill 2 & 3 Trash Screens			Greenfields UFG Mill 2 & 3 Fresh Feed Inclined Static Screen	Greenfields UFG Mill 2 & 3 Fresh Feed Inclined Static Screen	Greenfields UFG Mill 2 & 3 Fresh Feed Inclined Static Screen	Greenfields UFG Mill 2 & 3 Fresh Feed Inclined Static Screen				A
	Feed Stream							1	Northern Star	23	B
	Type							1	Northern Star	74	A
	Number of Screens per UFG Circuit	No.		1	1	1	1	1	Northern Star	23	A
	Screen Solids Feed Rate per Screen	P ₈₀ 9 µm	t/h	37	46	46	42	6	Calculated		
	Screen Feed Density	P ₈₀ 9 µm	% w/w	59.6	59.0	59.0	59.6	6	Calculated		
	Screen Feed Flowrate per Screen	P ₈₀ 9 µm	m³/h	33	41	41	37	6	Calculated		
	Screen Size	L x W	m	1,240 x 1.6	1,240 x 1.6	1,240 x 1.6	1,240 x 1.6	5	Vendor	77	B
	Screen Area per Screen		m²	2.84	2.84	2.84	2.84	5	Vendor	77	B
	Specific Screen Flux Rate		m³/m²/h	14.8	17.7	17.7	16.1	6	Calculated		
	Screen Aperture		mm	1.0	1.0	1.0	1.0	1	Northern Star	16	A
	Screen Material			Wedge Wire Stainless Steel	Wedge Wire Stainless Steel	Wedge Wire Stainless Steel	Wedge Wire Stainless Steel	5	Vendor	77	B
	Screen Spray Water		m³/h	9.0	9.0	9.0	9.0	5	Vendor	77	B
9.6	UFG Mill 2 & 3 Classification			Greenfields 100CVX	Greenfields 100CVX	Greenfields 100CVX	Greenfields 100CVX				A
	Cyclone Model			1	1	1	1	1	Northern Star	16	A
	Number of Cyclone Clusters per UFG Circuit			1	1	1	1	5	Vendor	54	A
	Number Installed Cyclones per Cluster	No.		24	24	24	24	5	Vendor	54	B
	Number of Cyclone Distributor Outlets Available	No.		24	24	24	24	5	Vendor	54	A
	Spare Outlets Available	No.		0	0	0	0	6	Calculated		
	Number of Operating Cyclones	No.		14	14	14	19	5	Vendor	54	B
	Number of Standby Cyclones	No.		10	10	10	5	6	Calculated		
	Cyclone Solids Feed Rate - per UFG circuit	P ₈₀ 9 µm	t/h	147.3	161.0	161.0	146.1	6	Calculated		
	Cyclone Feed Flow - per UFG circuit	P ₈₀ 9 µm	m³/h	319.0	348.5	348.5	315.1	6	Calculated		
	Cyclone Feed Solids Concentration		% w/w	32.8	32.8	32.8	32.9	5	Vendor	65	B
	Underflow Solids Concentration	P ₈₀ 9 µm	t/h	48.0	48.0	48.0	48.0	5	Vendor	65	B
	Underflow Solids - per UFG circuit	P ₈₀ 9 µm	t/h	109.9	115.0	115.0	104.4	6	Calculated		
	Overflow Solids Concentration	P ₈₀ 9 µm	% w/w	17.0	18.3	18.3	18.4	6	Calculated		
	Overflow Solids - per UFG circuit	P ₈₀ 9 µm	t/h	37.5	46.0	46.0	41.8	6	Calculated		
	Cyclone Circulating Load		%	293	250	250	250	5	Vendor	54	0
	Operating Pressure		kPa	247	247	247	248	5	Vendor	54	B
	Product Size P ₈₀		µm	9	9	9	9	1	Northern Star	5	A
9.7	Concentrate Pre-Leach Thickener (Brownfields)			Existing	Existing	Existing	Existing			Existing Pre-Leach Thickener 30-TH-31 Repurposed	
	Feed Streams			UFG Circuit 1, 2 & 3 Cyclone OI's	UFG Circuit 1, 2 & 3 Cyclone OI's	UFG Circuit 1, 2 & 3 Cyclone OI's	UFG Circuit 1, 2 & 3 Cyclone OI's	1	Northern Star	24	A
	Solids Feed Rate		t/h	75	104	104	84	6	Calculated		
	Feed Flowrate - Excl. Diluted Flocculant		m³/h	362	475	475	368	6	Calculated		
	Feed Solids Concentration		% w/w	17	18	18	18	6	Calculated		
	Underflow Solids Concentration		% w/w	55	55	55	55	1	Northern Star	115	1
	Calculated Flux Rate		t/m²/h	0.13	0.18	0.18	0.15	6	Calculated		
	Testwork Flux Rate		t/m²/h	0.25	0.25	0.25	0.25	1	Northern Star	103	A
	Thickener Area Calculated		m²	573	573	573	573	6	Calculated		
	Diameter	Existing	m	27	27	27	27	1	Northern Star	103	A
	Thickener Spray Water		m³/h	15	15	15	15	3	Assumed		C
10.0	CIL - FLOTATION CONCENTRATE										
10.1	General Circuit (Brownfields)			Existing CIL 2 & 3	Existing CIL 2 & 3	Existing CIL 2 & 3	Existing CIL 2 & 3				
	Solids Feed Rate	Fresh	t/h	75	104	104	84	6	Calculated		
	Feed Pulp Density	Range	% w/w	50 - 60	50 - 60	50 - 60	50 - 60	1	Northern Star	108	0
	Actual Feed Pulp Density	Total	% w/w	49	49	49	49	6	Calculated		
	Slurry Feed Flow Rate	Fresh	m³/h	77	107	107	86	6	Calculated		
	Slurry Feed Flow Rate	Total	m³/h	99	137	137	110	6	Calculated		
	Leach Gold Extraction		%	90	90	90	90	9	Agreed		2
	Adsorption Efficiency		%	99	99	99	99	1	Northern Star	44	B
10.2	Leaching (Brownfields)			Existing CIL 2 Concentrate Pre-Leach Thickener Underflow & Portion of Spent Eluate (CIL2/3, CIL4 & Gravity)	Existing CIL 2 Concentrate Pre-Leach Thickener Underflow & Portion of Spent Eluate (CIL2/3, CIL4 & Gravity)	Existing CIL 2 Concentrate Pre-Leach Thickener Underflow & Portion of Spent Eluate (CIL2/3, CIL4 & Gravity)	Existing CIL 2 Concentrate Pre-Leach Thickener Underflow & Portion of Spent Eluate (CIL2/3, CIL4 & Gravity)			CIL 2 repurposed for leach duty	
	Feed Streams							1	Northern Star	106	0
	Leach pH										
	30-TK-32A - 30-TK-32C			11.75 - 12.5	11.75 - 12.5	11.75 - 12.5	11.75 - 12.5	1	Northern Star	108	0
	30-TK-32D			11.5 - 12.5	11.5 - 12.5	11.5 - 12.5	11.5 - 12.5	1	Northern Star	108	0
	30-TK-32E			11 - 12.5	11 - 12.5	11 - 12.5	11 - 12.5	1	Northern Star	108	0
	Number of Tanks			7	7	7	7	1	Northern Star	114	1
	Effective Tank Capacity	Per Tank	m³	1,685	1,685	1,685	1,685	1	Northern Star	16	A
	Total Tank Capacity		m³	11,795	11,795	11,795	11,795	6	Calculated		
	Residence Time		h	119	86	107	107	6	Calculated		
	Minimum Residence Time		h	80	80	80	80	1	Northern Star	103	A
	Oxygen Addition Points			All Tanks	All Tanks	All Tanks	All Tanks	1	Northern Star	16	A
	Oxygen Addition Method			Side Lances on Tanks A to F and Oxygen Circulating Pump on Tanks A and B	Side Lances on Tanks A to F and Oxygen Circulating Pump on Tanks A and B	Side Lances on Tanks A to F and Oxygen Circulating Pump on Tanks A and B	Side Lances on Tanks A to F and Oxygen Circulating Pump on Tanks A and B	1	Northern Star	43	4
	Dissolved Oxygen Concentration	Target	ppm	10	10	10	10	1	Northern Star	16	A
	Cyanide Addition Points			All Tanks	All Tanks	All Tanks	All Tanks	1	Northern Star	43	A
	Cyanide Concentrations										
	30-TK-32A	Target	ppm	500	500	500	500	1	Northern Star	108	0
	30-TK-32B	Target	ppm	1,000	1,000	1,000	1,000	1	Northern Star	108	0
	30-TK-32C	Target	ppm	2,200	2,200	2,200	2,200	1	Northern Star	108	0
	30-TK-32D	Target	ppm	5,000	5,000	5,000	5,000	1	Northern Star	108	0
	30-TK-32E	Target	ppm	6,500	6,500	6,500	6,500	1	Northern Star	108	0
	Milk of Lime Addition Points			Leach Feed Box	Leach Feed Box	Leach Feed Box	Leach Feed Box	1	Northern Star	43	A
	Lead Nitrate Addition Points			Leach Feed Box	Leach Feed Box	Leach Feed Box	Leach Feed Box	1	Northern Star	43	A
10.3	Adsorption (Brownfields)			Existing CIL 3	Existing CIL 3	Existing CIL 3	Existing CIL 3			CIL 3 repurposed for adsorption duty	
	Number of Tanks			6	6	6	6	1	Northern Star	114	1
	Effective Tank Capacity	Per Tank	m³	1,685	1,685	1,685	1,685	1	Northern Star	16	A
	Total Adsorption Tank Live Volume		m³	10,110	10,110	10,110	10,110	6	Calculated		
	Residence Time		h	102	74	92	92	6	Calculated		
	Minimum Residence Time		h	60	60	60	60	1	Northern Star	103	A
	Total Leach & Adsorption Time		h	221	160	160	199	6	Calculated		
	Carbon Concentration	per tank	g/L	20	20	20	20	1	Northern Star	16	A
	Maximum Carbon Residence Time		days	14	14	14	14	1	Northern Star	103	A
	Carbon Movements per Day	Max	t/d	33	33	33	33	6	Calculated		
	Carbon In Circuit	Total	t	202.2	202.2	202.2	202.2	6	Calculated		
	Design Carbon Loading		g/t	4,490	4,490	4,490	4,490	1	Northern Star	16, 103	A
	Carbon Transfer Method			Recessed Impeller Pump	Recessed Impeller Pump	Recessed Impeller Pump	Recessed Impeller Pump	1	Northern Star	16	A
	Carbon Transfer Period	per cycle	h	4.0	4.0	4.0	4.0	6	Calculated		
	Carbon and Pulp Transfer Rate		m³/h	138	138	138	138	2	Primerio	119	4

AREA	PARAMETER	UNIT	DATA				REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV
				1	2	3	4				
			FIMISTON	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um	NEW CIRCUIT 28.35 MTPA THROUGHPUT NOMINAL GRADE P80 220um				
10.4	Loaded Carbon Recovery Screen & Transfer			Recessed Impeller Pump	Recessed Impeller Pump	Recessed Impeller Pump	Recessed Impeller Pump	2	Primero		
	Loaded Carbon Advance Method			275	275	275	275	2	Primero		1
	Loaded Carbon Pump & Pulp Forwarding Rate			2	2	2	2	2	Primero		1
	Loaded Carbon Transfer Time	Pump to Screen									1
	Feed Streams		Loaded Carbon Slurry from CIL3 Tank A (or Tank B)	Loaded Carbon Slurry from CIL3 Tank A (or Tank B)	Loaded Carbon Slurry from CIL3 Tank A (or Tank B)	Loaded Carbon Slurry from CIL3 Tank A (or Tank B)	1	Northern Star		As per KCGM-P3-EPC-COR-602	1
	Type		Linear Motion	Linear Motion	Linear Motion	Linear Motion	1	Northern Star		New Derrick Linear Motion Screen as per KCGM-P3-EPC-COR-602	1
	Number	No.	1	1	1	1	1	Northern Star		New Derrick Linear Motion Screen as per KCGM-P3-EPC-COR-602	1
	Feed Rate	m³/h	275	275	275	275	6	Calculated			
	Area	m²	3.80	3.80	3.80	3.80	5	Vendor	117		2
	Deck Type		Derrick Polyweb Urethane	Derrick Polyweb Urethane	Derrick Polyweb Urethane	Derrick Polyweb Urethane	5	Vendor	117		1
	Aperture	mm	0.8 x 12.8	0.8 x 12.8	0.8 x 12.8	0.8 x 12.8	5	Vendor	117		1
	Total Spray Water Addition	m³/h	10	10	10	10	3	Assumed			1
	Loaded Carbon Moisture	% w/w	40	40	40	40	1	Northern Star	4	Screen Oversize From Transfer Hopper	1
	Loaded Carbon Transfer Moisture	Maximum % w/w	55	55	55	55	1	Northern Star	4		1
	Loaded Carbon Transfer Hopper Capacity	Minimum m³	23	23	23	23	2	Primero			4
	Loaded Carbon Transfer Time	Hopper to Column	60	60	60	60	2	Primero			1
10.5	Interstage Carbon Screens (Brownfields)			Existing CIL 3	Existing CIL 3	Existing CIL 3	Existing CIL 3				
	Type		Mechanically Swept Pumped Screen	Mechanically Swept Pumped Screen	Mechanically Swept Pumped Screen	Mechanically Swept Pumped Screen	1	Northern Star	16		A
	Supplier & Model		Kemix MPSP 1200	Kemix MPSP 1200	Kemix MPSP 1200	Kemix MPSP 1200	1	Northern Star	16		A
	Number	per tank	1	1	1	1	1	Northern Star	16		A
	Screen Feed Rate - per Screen	w/o carbon advance	102	140	140	113	6	Calculated			
		w/ carbon advance	240	278	278	251	6	Calculated			
	Area Selected	Selected	12.0	12.0	12.0	12.0	1	Northern Star	16		A
	Specific Screen Flux Rate	w/ carbon advance	20.0	23.1	23.2	20.9	6	Calculated			
		Maximum	100	100	100	100	1	Northern Star	16		B
	Screen Material		Wedge Wire Stainless Steel	Wedge Wire Stainless Steel	Wedge Wire Stainless Steel	Wedge Wire Stainless Steel	1	Northern Star	102		A
	Screen Aperture	mm	1.0	1.0	1.0	1.0	1	Northern Star	16		A
10.6	CIL 3 Tails Hopper										
	Feed Flowrate	m³/h	106	144	144	118	6	Calculated			
	Discharge Destination		CIL 4 Feed Distribution Box	CIL 4 Feed Distribution Box	CIL 4 Feed Distribution Box	CIL 4 Feed Distribution Box	1	Northern Star	43		0
	Cyanide Concentration	ppm	750	750	750	750	1	Northern Star	108		0
11.0 FLOTATION TAILINGS THICKENING											
11.1	Flotation Tailings Thickener			Scavenger Flotation Tails Cleaner-Scavenger Flotation Tails	Scavenger Flotation Tails Cleaner-Scavenger Flotation Tails	Scavenger Flotation Tails Cleaner-Scavenger Flotation Tails	Scavenger Flotation Tails Cleaner-Scavenger Flotation Tails	1	Northern Star	16	NSR Supplied
	Feed Stream										
	Solids Feed Rate	th	3,071	2,858	3,042	3,411	6	Calculated			
	Feed Flowrate - Excl. Diluted Flocculant	m³/h	6,871	6,492	6,894	6,187	6	Calculated			
	Feed Solids Concentration - Excl. Diluted Flocculant	% w/w	33.0	32.6	32.7	38.9	6	Calculated			
	Underflow Solids Concentration	Nominal % w/w	55.9	56.0	56.0	55.9	6	Calculated		Achieve 55 % solids concentration in flotation tails CIL circuit post dilution (i.e. total feed % solids)	A
	Underflow Solids Concentration Range	% w/w	50-62	50-62	50-62	50-62	1	Northern Star	23		A
	Testwork Flux Rate	t/m²/h	1.50	1.50	1.50	1.50	1	Northern Star	23		A
	Operating Flux Rate Range	t/m²/h	1.05 - 1.27	1.05 - 1.27	1.05 - 1.27	1.05 - 1.27	1	Northern Star	16		A
	Nominal Flux Rate	t/m²/h	1.09	1.01	1.08	1.21	6	Calculated			
	Thickener Area Calculated	m²	2,827	2,827	2,827	2,827	6	Calculated			
	Diameter Selected	m	60	60	60	60	1	Northern Star	16		A
	Thickener Underflow Destination(s)		CIL 4 Feed Distribution Box	CIL 4 Feed Distribution Box	CIL 4 Feed Distribution Box	CIL 4 Feed Distribution Box	1	Northern Star	43		A
	Number of Underflow Discharge Pumps		2	2	2	2	1	Northern Star	16		B
	Pump Configuration		Duty/Standby	Duty/Standby	Duty/Standby	Duty/Standby	1	Northern Star	43		A
	Total Thickener Underflow	m³/h	3,329	3,087	3,289	3,701	6	Calculated			
12.0 CIL - FLOTATION TAILINGS											
12.1	General Circuit			CIL 4	CIL 4	CIL 4	CIL 4				
	Feed Streams		Flotation Tailings Thickener U/F, CIL 3 Tailings & Portion of Spent Eluate (CIL2/3, CIL4 & Gravity)	Flotation Tailings Thickener U/F, CIL 3 Tailings & Portion of Spent Eluate (CIL2/3, CIL4 & Gravity)	Flotation Tailings Thickener U/F, CIL 3 Tailings & Portion of Spent Eluate (CIL2/3, CIL4 & Gravity)	Flotation Tailings Thickener U/F, CIL 3 Tailings & Portion of Spent Eluate (CIL2/3, CIL4 & Gravity)	1	Northern Star	106		0
	Number of Trains		1	1	1	1	1	Northern Star	43		A
	Fresh Solids Feed Flow Rate	th	3,151	2,969	3,153	3,500	6	Calculated		Flotation Tailings Thickener U/F & CIL 3 Tailings	
	Total Solids Feed Flow Rate	th	3,164	2,982	3,166	3,515	6	Calculated			
	Fresh Slurry Feed Flow Rate	m³/h	3,438	3,234	3,437	3,822	6	Calculated		Flotation Tailings Thickener U/F & CIL 3 Tailings	
	Total Slurry Feed Flow Rate	m³/h	3,511	3,305	3,510	3,900	6	Calculated			
	Target Feed Pulp Density	Target % w/w	55	55	55	55	1	Northern Star	16, 103	Post dilution (i.e. total feed % solids)	A
	Maximum Feed Pulp Density	% w/w	60	60	60	60	1	Northern Star	43	Post dilution (i.e. total feed % solids)	A
	Feed Pulp Density	Actual % w/w	55	55	55	55	6	Calculated		Post dilution (i.e. total feed % solids)	
	Leach Gold Extraction	%	38.0	38.0	38.0	38.0	1	Northern Star	44		B
	Adsorption Efficiency	%	88.0	88.0	88.0	88.0	1	Northern Star	44	Range 84 - 92%	B
12.2	Leaching			CIL 4	CIL 4	CIL 4	CIL 4				
	Leach pH		8.8 - 9.3	8.8 - 9.3	8.8 - 9.3	8.8 - 9.3	1	Northern Star	108		0
	F21-TK-411		2	2	2	2	1	Northern Star	43		A
	Number of Tanks	per train	2	2	2	2	1	Northern Star	43		B
	Live Tank Capacity	m³	5,600	5,600	5,600	5,600	1	Northern Star	43		
	Total Tank Capacity	m³	11,200	11,200	11,200	11,200	6	Calculated			
	Residence Time	h	3.2	3.4	3.2	2.9	6	Calculated			
	Oxygen Addition Points		All Leach Tanks	All Leach Tanks	All Leach Tanks	All Leach Tanks	1	Northern Star	43		A
	Oxygen Addition Method		Side Lances and Circulating Pumps on both Leach Tanks	Side Lances and Circulating Pumps on both Leach Tanks	Side Lances and Circulating Pumps on both Leach Tanks	Side Lances and Circulating Pumps on both Leach Tanks	1	Northern Star	43		A
	Dissolved Oxygen Concentration	Target	15	15	15	15	1	Northern Star	108		0
	Cyanide Addition Points		Feed Box and Tank 2	Feed Box and Tank 2	Feed Box and Tank 2	Feed Box and Tank 2	1	Northern Star	43		B
	Cyanide Concentration F21-TK-411	ppm	175	175	175	175	1	Northern Star	108		0
	Milk of Lime Addition Points		Leach Feed Box	Leach Feed Box	Leach Feed Box	Leach Feed Box	1	Northern Star	43		A
12.3	Adsorption			CIL 4	CIL 4	CIL 4	CIL 4				
	Number of Tanks	per train	6	6	6	6	1	Northern Star	43	Provision in the layout for a future, 7th adsorption tank	A
	Live Tank Capacity	m³	5,600	5,600	5,600	5,600	1	Northern Star	43		B
	Total Adsorption Tank Live Volume	m³	33,600	33,600	33,600	33,600	6	Calculated			
	Residence Time	h	9.6	10	10	9	6	Calculated			
	Total Leach & Adsorption Time	h	12.8	13.6	12.8	11.5	6	Calculated		2 Leaching & 6 Adsorption Tanks will not achieve the 14 hours total residence time. Client is aware and has advised to maintain 8 tanks with the 9th as a future.	
	Oxygen Addition Points		First Two Adsorption Tanks	First Two Adsorption Tanks	First Two Adsorption Tanks	First Two Adsorption Tanks	1	Northern Star	43		A
	Oxygen Addition Method		Side Lances	Side Lances	Side Lances	Side Lances	1	Northern Star	43		A
	Carbon Concentration - Duty	per tank	12.5	12.5	12.5	12.5	1	Northern Star	109		1
	Carbon Concentration Used for Carbon Transfer	g/L	12.5	12.5	12.5	12.5	1	Northern Star	109		1
	Carbon Movements per Day	Max t/d	30	30	30	30	6	Calculated			
	Carbon Residence Time	d	14	14	14	14	1	Northern Star	103		A
	Carbon In Circuit	t	269	269	269	269	6	Calculated			
	@ 8 g/L	t	336	336	336	336	6	Calculated			
	@ 10 g/L	t	420	420	420	420	6	Calculated			
	@ 12.5 g/L	t	461	461	461	461	1	Northern Star	16, 103		A
	Nominal Carbon Loading	g/t	332 - 579	332 - 579	332 - 579	332 - 579	1	Northern Star	16, 103		A
Carbon Loading Range	g/t										
Carbon Transfer Method		Recessed Impeller Pump	Recessed Impeller Pump	Recessed Impeller Pump	Recessed Impeller Pump	2	Primero			0	
Carbon Transfer Period	h	5.5	5.5	5.5	5.5	6	Calculated				
Carbon and Pulp Transfer Rate	m³/h	220	220	220	220	2	Primero			0	
Loaded Carbon Advance Method		Recessed Impeller Pump	Recessed Impeller Pump	Recessed Impeller Pump	Recessed Impeller Pump	2	Primero			0	
Loaded Carbon Pump & Pulp Forwarding Rate	m³/h	280	280	280	280	2	Primero			0	
Loaded Carbon Transfer Time	h	4.3	4.3	4.3	4.3	6	Calculated				
12.4	Loaded Carbon Recovery Screen			CIL 4	CIL 4	CIL 4	CIL 4				
	Feed Streams		First Adsorption Tank	First Adsorption Tank							

AREA	PARAMETER	UNIT	DATA				REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV
			1	2	3	4					
			FIMISTON	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um					
14.3	Acid Washing										
	Acid Washes Per Week	No. t		12			6	Calculated			
	Acid Wash Column Capacity			15			1	Northern Star	8	Combined Acid Wash & Elution column	A
	Acid Type			Hydrochloric Acid			1	Northern Star	103		A
	Acid Dosing Method			Pumped			1	Northern Star	103		A
	Acid Concentration	% w/v HCl		3			1	Northern Star	103		A
	Acid Injection Time	min		17			1	Northern Star	103		A
	Acid Soak Time	min		60			1	Northern Star	103		A
	Acid Water Rinse Time	min		120			1	Northern Star	103		A
	Total Acid Wash Solution Flowrate	BV/h		2			1	Northern Star	8		A
	Total Acid Wash Solution Flowrate	m³/h		63.8			6	Calculated			
	Dilute Acid Flowrate	BV		0.6			6	Calculated			
	Dilute Acid Solution Used	m³		63.8			6	Calculated			
	Neat Acid Solution Used	m³		18.085			6	Calculated			
	Waste Acid Solution Destination			1.5			6	Calculated			
	Acid Wash Temperature	°C		Final Tails Tank Ambient			1	Northern Star	43		B
							1	Northern Star	44		A
14.4	Water Rinse										
	Water Flowrate	m³/h		63.8			6	Calculated		Based on 2 BV/h	
	Acid Water Rinse Time	min		120			1	Northern Star	103		A
	Volume Water Used	m³		128			6	Calculated			
	No. Bed Volumes	BV		4			1	Northern Star	8, 103		
	Solution Destination			Final Tails Tank Ambient			1	Northern Star	43		B
	Acid Rinse Temperature	°C		Scheme Water			1	Northern Star	44		A
	Rinse Water Source						1	Northern Star	103		A
14.5	Elution										
	Elution Type			Split AARL			1	Northern Star	43		A
	Required Elutions Per Week	No. t		12			1	Northern Star	8, 43		B
	Elution Column Capacity			15			1	Northern Star	8, 43		A
	Carbon Bed Volume	m³		31.9			6	Calculated			
	Elution Column Operating Pressure	Top of Column kPa(g)		340			2	Primero		To prevent flashing (boiling) of solution inside column.	B
14.6	Pre-treatment (with Lean Eluate)										
	Cycle Time	min		30			2	Primero			A
	Solution Flowrate	BV/h		2			1	Northern Star	8		A
	Temperature	°C		63.8			6	Calculated			
	No. of Bed Volumes			95			2	Primero			A
	Pre-soak Solution Volume	m³		1.0			6	Calculated		5 BVs of Pregnant Eluate required by NSR (Source 8)	
	Column Discharge Destination			31.9			6	Calculated			
	Solution Strength	% w/v NaCN		Lean Eluate Tank			8	Industry			B
				1.7			1	Northern Star	103		A
	Sodium Cyanide Consumption	% w/v NaOH		2			1	Northern Star	103		A
	Caustic Consumption	kg/pre-treat	Neat Solution	543			6	Calculated			
	Lean Eluate Solution Make-up Per Mix	m³/pre-treat	Neat Solution	1.51			6	Calculated			
		m³/mix		638			6	Calculated			
				0.85			6	Calculated			
				29.5			6	Calculated			
14.7	Elution (with Lean Eluate)										
	Elution Cycle Time	min		150			2	Primero			A
	Elution Solution Flowrate	BV/h		2			1	Northern Star	8		A
	Eluate Consumed	m³/h		63.8			6	Calculated			
	Elution Temperature	°C		5.0			6	Calculated		5 BVs of Pregnant Eluate required by NSR (Source 15)	
	Column Discharge Destination			159.6			6	Calculated			
				130			2	Primero			A
				CIL4 Eluate Circulation Tanks			8	Industry			A
14.8	Elution & Cooling (with Scheme Water)										
	Cycle Time - Scheme Water (with heaters)	min		90			6	Calculated			
	Elution Temperature	°C		130			2	Primero			C
	Water Flowrate	BV/h		2			1	Northern Star	8		A
	Water Consumed	m³/h		63.8			6	Calculated			
		BV		3.0			6	Calculated			
		m³		95.7			6	Calculated			
	Cycle Time - Scheme Water (no heaters, cooling)	min		60			2	Primero			A
	Elution Cookdown Temperature	°C		-80			2	Primero			A
	Water Flowrate	BV/h		2			1	Northern Star	8		A
	Water Consumed	m³/h		63.8			6	Calculated			
		BV		2.0			6	Calculated			
		m³		63.8			6	Calculated			
	Total Cycle Time	min		150			2	Primero			A
	Water Flowrate	BV/h		2			1	Northern Star	8		A
	Total Water Consumed	m³/h		63.8			6	Calculated			
		BV		5.0			6	Calculated			
		m³		159.6			6	Calculated		Total of 5 BVs of Lean Eluate required by NSR (Source 8)	
	Column Discharge Destination			CIL4 Eluate Circulation Tanks and Lean Eluate Tank			8	Industry		When the CIL4 eluate circulation tank is full, discharge switches to the lean eluate tank	A
	Carbon Transfer In	min		30.0			6	Calculated			
	Total Acid Time (including rinsing)	h		0.5			6	Calculated			
		min		197			6	Calculated			
		h		3.3			6	Calculated			
14.9	Heating										
	Elution Heater Type			LPG Fired Elution Heater			1	Northern Star	43		A
14.10	Lean Eluate Tank										
	Total Volume of Elution Eluate Per Strip	Pregnant m³		160			6	Calculated			
	Number of Lean Eluate Tanks			1			1	Northern Star	43		A
14.11	Elution Water Tank										
	Total Live Volume Lean Eluate Tank	Minimum m³		160			6	Calculated			
14.12	Eluate Expansion Tank										
	Number of Eluate Expansion Tanks			1			2	Primero			B
	Total Live Volume Eluate Expansion Tank	m³		3.5			6	Calculated			
14.13	Total Water Consumption Per Strip										
	Carbon Transfer	Scheme m³/strip		8			6	Calculated			
	Acid Wash	Scheme m³/strip		17			6	Calculated			
	Water Rinse	Scheme m³/strip		128			6	Calculated			
	Water Elution & Cooling	Scheme m³/strip		160			6	Calculated			
	Carbon Eduction (Elution to Regen)	Scheme m³/strip		60			6	Calculated			
	Carbon Quench to Sizing Screen	Scheme m³/strip		60			6	Calculated			
	Total Per Strip	Scheme m³/strip		432			6	Calculated			
14.14	Elution Electrowinning										
	Number Circuits			1			2	Primero			A
	Number of Cells Per Circuit			2			5	Vendor	48	2 off Split (2 in 1) Clean-in-Place cells	A
	Cell Configuration			Parallel - 2 off Split (2 in 1)			5	Vendor	48		B
	Cell Size	m x m		1.00 x 1.00			5	Vendor	48		A
	Effective Cross Sectional Area	m²		1.00			6	Calculated			
	Pregnant Eluate Volume	m³		160			6	Calculated			
	Pregnant Eluate Grade	g Au/m³		52.6			6	Calculated			
	Number of Eluate Volume Passes			5.64			5	Vendor	48		A
	Electrowinning Feed Flowrate per Circuit	Total m³/h		90.0			6	Calculated			
	Electrowinning Feed Flowrate per Split Cell	per Split Cell m³/h		45.0			6	Calculated			
	Electrowinning Feed Flowrate per Individual Cell	per Individual Cell m³/h		22.5			6	Calculated			
	Maximum Cell Linear Velocity	mm/min		0.450			2	Primero	48		A
	Actual Cell Linear Velocity	mm/min		0.375			5	Vendor	48		A
	Electrowinning Time	h		10			1	Northern Star	103		A
	Cathode Type			Stainless Steel 316 Mesh			5	Vendor	48		A
	Number Cathodes per Individual Cell	No.		11			5	Vendor	48		A
	Number Cathodes per Split Cell	No.		22			5	Vendor	48		A
	Number of Rectifiers	No.		4			5	Vendor	48		B
	Rectifier Size	A		2,500			5	Vendor	48		B
	Target Barren Solution Grade	ppm Au		<5			2	Primero	8, 103		A
	Barren Solution Pumping Time To Leach	h		8			2	Primero			B
	Cleaning Method			Clean in Place			1	Northern Star	8		A
15.0	GOLDROOM										
15.1	Goldroom Operation (Brownfields)			Existing							
	Feed Sources			Flotation Con EW product (CIL2/3) & Flotation Tails EW product (CIL 4)			1	Northern Star	43		A
	Sludge Filter Operating Days	days/week		2			1	Northern Star	16		A
	Sludge Dewatering Cone Capacity	m³		1.3			1	Northern Star	16		A
	Sludge Filter Make			TEFSA			1	Northern Star	16		A
	Sludge Filter Model			HPLE			1	Northern Star	16	470x470mm plate filter, 6 bar design pressure	A
	Mercury Retort Oven Operating Days	days/week		2			1	Northern Star	16		A
	Mercury Retort Oven Make			Locchead			1	Northern Star	16		A
	Mercury Retort Oven Model			F66OV66			1	Northern Star	16	20 ft³ electric powered mercury retort	A
	Calcine per week	kg		800 - 1,200			1	Northern Star	16	Existing, Calcine logs, Max Capacity 1,600 kg per cycle	A
	Smelting Furnace Operating Days	days/week		2			1	Northern Star	16		A
	Furnace Make			Furnace Engineering			1	Northern Star	16		A
	Furnace Model			Westofen KBU 80/H			1	Northern Star	16	2000 L x 3000 D x 1710 H, Weishaupt G3-1a burner	A
	Furnace Extraction Fan			75 kW / 1,480 RPM			1	Northern Star	16	IP55, 415V with 315xSSPB motor pulley and 212xSSPB fan pulley	A
	Furnace Scrubber			Metalcraft SN50247/1			1	Northern Star	16		A
	Smelts per week			6			1	Northern Star	16	Two downloads per week, three pours per day	A

AREA	PARAMETER	UNIT	DATA				REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV
			1	2	3	4					
			FIMISTON	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um					
16.0	CARBON REGENERATION - FLOTATION CONCENTRATE										
16.1	Barren Carbon Transfer (Brownfields)			Existing							
	Transfer Water Source			Carbon Transfer Water (Scheme Water)			1	Northern Star	16, 103		A
	Carbon Transfer Time	min		30			1	Northern Star	16, 103		A
	Carbon Transfer Concentration	% w/w		20			1	Northern Star	57		B
	Transfer Water Flowrate	m³/h	Each	88			6	Calculated		Based off 11t elution column	
	Transfer Water Required	m³	Each	44			6	Calculated		Based off 11t elution column	
16.2	Regeneration Kiln Dewatering Screens (Brownfields)			Existing							
	Type			Horizontal Vibrating Screen			1	Northern Star	16, 103		A
	Number	No.		3			1	Northern Star	16, 103	1x Per Regeneration Kiln	A
	Specific Screen Flux Rate	t/m²/h	Max Recommended	4			1	Northern Star	16, 103		A
	Selected Size	m	Width	1.2			1	Northern Star	16, 103		A
		m	Length	2.4			1	Northern Star	16, 103		A
	Area	m²		2.9			6	Calculated			
	Feed % Solids	% w/w		20			1	Northern Star	57		B
	% Mass Split to Oversize	%		100			2	Primerio			B
	Oversize Moisture %	% w/w		60			1	Northern Star	57		B
16.3	Carbon Regeneration Kiln (Brownfields)			Existing							
	Regeneration Kiln Type			Horizontal			1	Northern Star	16, 103		A
	Regeneration Kiln Model			Metso RK600/650HR			1	Northern Star	16, 103		A
	Number			3			1	Northern Star	16, 103		A
	Eluted Carbon to be Regenerated	%		100			1	Northern Star	16, 103		A
	Regenerated Carbon per Batch	t		11			6	Calculated			
	Temperature	°C		750			1	Northern Star	16, 103		A
	Required Kiln Operating time	h/batch		20			1	Northern Star	100		A
	Capacity - Nominal	kg/h		550			6	Calculated			
	Capacity - Design	kg/h		650			1	Northern Star	100		A
16.4	Carbon Quench Tank (Brownfields)			Existing							
	Carbon Quench Tank Volume	m³	Each	12			1	Northern Star	16, 103	3 off Quench Tanks	A
	Carbon Transfer Time	min	Each	30			1	Northern Star	16, 103		A
	Carbon Transfer Concentration	% w/w		20			1	Northern Star	57		B
	Transfer Water Flowrate	m³/h	Each	88			6	Calculated		Based off 11t elution column	
	Transfer Water			Carbon Transfer Water (Scheme Water)			1	Northern Star	16, 103		B
	Transfer Water Required	m³	Each	44			6	Calculated			
16.5	CIL 3 Barren Carbon Sizing Screen			New							
	Type			Horizontal Vibrating Screen			1	Northern Star	100		A
	Number	No.		1			1	Northern Star	100		A
	Feed Rate	m³/h		98			6	Calculated		Based off 1 x 11t elution columns	
	Feed Rate	t/h		22			6	Calculated		Based off 1 x 11t elution columns	
	Selected Size	m	Width	1.53			5	Vendor	84		0
		m	Length	3.66			5	Vendor	84		0
	Area	m²		5.60			6	Calculated			
	Specific Screen Flux Rate	m³/m²/h	Actual	17.5			6	Calculated		w/o screen spray water	
	Deck Type			Polyurethane			1	Northern Star	100		A
	Aperture	mm		1.25 x 18.5			5	Vendor	84		1
	% Mass Split to Oversize	%		100			2	Primerio	84		B
	Oversize Moisture %	% w/w		40			1	Northern Star	4		B
	Screen Spray Water	m³/h		13			5	Vendor			0
16.6	Regeneration Kiln Off-Gas Treatment (Brownfields)			Existing							
	Venturi Scrubber			1			1	Northern Star	16, 103		A
	Number			375			1	Northern Star	16, 103		A
	Design Dust Load	mg/Nm³		546			1	Northern Star	16, 103		A
	Design Mercury	g/h		831			1	Northern Star	16, 103		A
	Design VOC	g/h		250			1	Northern Star	16, 103		A
	Design Gas Temperature	°C		Process			1	Northern Star	16, 103		A
	Water Type										
	Packed Bed Wet Scrubber			Existing							
	Number			1			1	Northern Star	16, 103		A
	Design Gas Flowrate	Nm³/h		6,481			1	Northern Star	16, 103		A
	Design Gas Temperature	°C		80			1	Northern Star	16, 103		A
	Water Type			Process			1	Northern Star	16, 103		A
	Mist Eliminator			Existing							
	Number			1			1	Northern Star	16, 103		A
	Design Gas Flowrate	Nm³/h		3,916			1	Northern Star	16, 103		A
	Design Gas Temperature	°C		58			1	Northern Star	16, 103		A
	Water Type			Scheme			1	Northern Star	16, 103		A
	Regenerative Thermal Oxidiser			Existing							
	Number			1			1	Northern Star	16, 103		A
	Design Gas Flowrate	Nm³/h		4,132			1	Northern Star	16, 103		A
	Design Gas Temperature	°C		58			1	Northern Star	16, 103		A
	Carbon Filters			Existing							
	Number			2 (Duty / Standby)			1	Northern Star	16, 103		A
	Design Gas Flowrate	Nm³/h		4,503			1	Northern Star	16, 103		A
	Design Gas Temperature	°C		80			1	Northern Star	16, 103		A
	Design Mercury Loading	g/h		387			1	Northern Star	16, 103		A
	Stack Emission Limits										
	Mercury	ppmv		0.20			1	Northern Star	16, 103	Fimiston ERP Commissioning Report	A
	VOC	ppmv		3.2			1	Northern Star	16, 103		A
17.0	CARBON REGENERATION - FLOTATION TAILINGS										
17.1	Barren Carbon Transfer			New - Flotation Tailings							
	Transfer Water Source			Carbon Transfer Water (Scheme Water)			1	Northern Star	103		B
	Carbon Transfer Time	min		53			2	Primerio			0
	Carbon Transfer Concentration	%w/w		20			1	Northern Star	57		B
	Transfer Water Flowrate	m³/h		67			6	Calculated		Based off 1x 15t carbon batch	
	Transfer Water Required	m³		60			6	Calculated		Based off 1x 15t carbon batch	
17.2	Regeneration Kiln Dewatering Screen			New							
	Type			Horizontal Vibrating Screen			5	Vendor	84		A
	Number	No.		1			1	Northern Star	103	One elution column	A
	Feed Flow Rate	m³/h		75			6	Calculated			
	Solids Feed Rate	t/h		17			6	Calculated			
	Selected Size	m	Width	1.53			5	Vendor	84		0
		m	Length	3.66			5	Vendor	84		0
	Area	m²		5.6			6	Calculated			
	Specific Screen Flux Rate	t/m²/h	Maximum	4			1	Northern Star	103		B
		m³/m²/h	Actual	13.4			6	Calculated			
		t/m²/h	Actual	3.0			6	Calculated			
	Deck Type			Polyurethane			5	Vendor	84		B
	Aperture	mm		1.2 x 18.5			5	Vendor	84		0
	Feed % Solids	% w/w		20			1	Northern Star	57		B
	% Mass Split to Oversize	%		100			2	Primerio			B
	Oversize Moisture %	% w/w	Nominal	40			2	Primerio	52		B
		% w/w	Maximum	60			1	Northern Star	57		B
	Screen Spray Water	m³/h		13			5	Vendor			0
17.3	Carbon Regeneration Kiln			New							
	Reactivation Kiln Type			Horizontal - Indirect Fired, Electric			1	Northern Star	103		B
	Number			1			1	Northern Star	103		A
	Regenerated Carbon per Batch	t		30			6	Calculated		2 strips per day from 15t elution column	
	Regen Kiln Feed Hopper Capacity	%	% of full column size	200			1	Northern Star	103		B
	Regen Kiln Feed Hopper Capacity	t		30			6	Calculated			
	Required Kiln Operating time	h/batch	Nominal	20			8	Industry			A
	Capacity - Nominal	kg/h		1,650			1	Northern Star	43		A
	Operating Temperature	°C		650 - 750			1	Northern Star	102		A
	Design Operating Temperature	°C		750			1	Northern Star	103		
	Fraction of Carbon Regenerated	%		100			1	Northern Star	103		
	Regenerated Carbon Mercury Grade	Hg ppm	Nominal	1.3			1	Northern Star	16		B
		Hg ppm	Range	0 - 4			1	Northern Star	16		B
17.4	Carbon Quench Tank			New							
	Carbon Quench Tank Volume	m³	Minimum	20			1	Northern Star			A
	Carbon Transfer Time	min		53			2	Primerio			0
	Carbon Transfer Concentration	% w/w		20			2	Primerio			C
	Transfer Water Flowrate	m³/h		67.3			6	Calculated		Based off one 15t elution column	
	Transfer Water			Quench Hopper filled with Scheme Water			1	Northern Star	100		0
	Transfer Water Required	m³		Carbon Transfer completed with Raw Water			6	Calculated		Based off one 15t elution column	
17.5	CIL 4 Carbon Sizing Screen			New							
	Type			Horizontal Vibrating Screen			1	Northern Star	100		A
	Number	No.		1			1	Northern Star	103	Single CIL4 Train	A
	Feed Flow Rate	m³/h		75			6	Calculated			
	Solids Feed Rate	t/h		17			6	Calculated			
	Selected Size	m	Width	1.53			5	Vendor	84		0
		m	Length	3.66			5	Vendor	84		0
	Area	m²		5.6			6	Calculated			
	Specific Screen Flux Rate	m³/m²/h	Maximum	35.0			1	Northern Star	103		B
		t/m²/h	Actual	3.0			6	Calculated			
		m³/m²/h	Actual	13.4			6	Calculated		w/o screen spray water	
	Deck Type			Polyurethane			5	Vendor	84		B
	Aperture	mm		1.25 x 18.5			5	Vendor	84		1
	% Mass Split to Oversize	%		100			2	Primerio	84		B
	Oversize Moisture %	% w/w		40			1	Northern Star	4		B
	Screen Spray Water	m³/h		13			5	Vendor			0

AREA	PARAMETER		UNIT	DATA				REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV	
				1	2	3	4						
				FIMISTON	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um						
17.6	Regeneration Kiln Off-Gas Treatment				New								
	Mercury Recovery	Target	%		>92			1	Northern Star	43		B	
	Design Mercury	New - Range	g/t		20 - 72			1	Northern Star	16	Brownfields barren carbon mercury grade CIL2/3 20th - 80th Percentile (2017 - 2023).	B	
		New - Design	g/h		108			6	Calculated		Based on the upper limit of brownfields barren carbon mercury grade. 72g/t for 1,500 kg carbon/h through regen kiln		
	VOC	New	g/h		639			6	Calculated		Based on brownfields design VOC of existing Venturi Scrubber. Assumes 3x existing Carbon Regen Kilns running at design throughput. Estimate only - stack emission criteria used as basis of design		
		New - Nominal	°C		316			5	Vendor	52		B	
	Gas Temperature	New - Design	°C		400			5	Vendor	52		1	
		New - Nominal	Nm³/h		1,335			6	Calculated	52	Based on 40% moisture CIL4 Carbon Regeneration Kiln feed	B	
	Gas Flow	New - Design	Nm³/h		3,003			6	Calculated	52	Based on 60% moisture CIL4 Carbon Regeneration Kiln feed	B	
	Carbon Fines Knock-Out Box												
	Number				1			5	Vendor	52		A	
	Dimensions		m		1.22 x 1.8			5	Vendor	52		1	
	Water Type	Sprays			Scheme			5	Vendor	52		A	
	Water Flowrate		m³/h		8.39			3	Assumed		Sprays used for maintenance only. Not a continuous consumption.	4	
	Kiln Off-Gas Condenser												
	Number				1			5	Vendor	52		A	
	Cooler Type				Shell and Tube			5	Vendor	52		1	
	Water Type				Cooling Water (Scheme)			5	Vendor	52		A	
	Kiln Mist Eliminator				New								
	Number				1			5	Vendor	52		A	
	Dimensions		m		1.22 x 1.8			5	Vendor	52		1	
	Water Type	Sprays			Scheme			5	Vendor	52		A	
	Water Flowrate		m³/h		8.39			3	Assumed		Sprays used for maintenance only. Not a continuous consumption.	4	
	Kiln Duct Heater												
	Number				1			5	Vendor	52		A	
	Power		kW		7.5			5	Vendor	52		1	
	Kiln Adsorption Vessel												
	Number				1			5	Vendor	52		A	
	Type				SIC Bed - CGS Dual Banjo			5	Vendor	52		B	
	Dimensions		m		1.98 x 2.7			5	Vendor	52		1	
	Carbon Type & Size		mm		3mm Mersorb HT			5	Vendor	52		1	
	Kiln Exhaust Fan												
	Number				1			5	Vendor	52		A	
	Design Duty		Am³/h		3,003			5	Vendor	52	At normal conditions of: 1 atm and 20 deg C	1	
	Power		kW		12.5			5	Vendor	52		1	
	Kiln Discharge Stack												
	Number				1			1	Northern Star	52		A	
	Height		m		35			1	Northern Star	52		B	
	Stack Emission Limits												
	Mercury	Maximum	ppmv		0.20			1	Northern Star	43	At standard temperature and pressure	A	
			micrograms/m³		1,800			1	Northern Star	52		B	
	Mercury	Vendor System	ppmv		0.027			6	Calculated			A	
	VOC	Maximum	ppmv		3.20			1	Northern Star	43		A	
18.0 CYANIDE DESTRUCT													
	Feed Streams	1			Kaltails Decant	Kaltails Decant	Kaltails Decant	Kaltails Decant	1	Northern Star	35	TBC with updated drawings by others	2
		2			Gidjji Decant & FIM I Decant	Gidjji Decant & FIM I Decant	Gidjji Decant & FIM I Decant	Gidjji Decant & FIM I Decant	1	Northern Star	35	TBC with updated drawings by others	2
		3			Decant Dam #3 & #4	Decant Dam #3 & #4	Decant Dam #3 & #4	Decant Dam #3 & #4	1	Northern Star	35	TBC with updated drawings by others	2
	Decant Water Recovered from Tailings	% of Total Tailings	%		49	49	49	49	1	Northern Star	35		A
		Water											
	Decant Water from Kaltails		m³/h		728	728	728	728	1	Northern Star	44		B
	Total Feed Flow to Cyanide Water Tank		m³/h		1,940	1,972	1,941	2,071	6	Calculated			
	Method				Hydrogen Peroxide	Hydrogen Peroxide	Hydrogen Peroxide	Hydrogen Peroxide	1	Northern Star	35		A
	Residual WAD Cyanide Target	Stage 2			<1.5	<1.5	<1.5	<1.5	1	Northern Star	35		A
	Dosing Points				Inline Mixers to each Decant Water line (3 off)	Inline Mixers to each Decant Water line (3 off)	Inline Mixers to each Decant Water line (3 off)	Inline Mixers to each Decant Water line (3 off)	1	Northern Star	35		2
	Dosing Points	Location			3	3	3	3	1	Northern Star	35		2
	Dosing Method	Number							1	Northern Star	35	Standby pump acts as a lag pump to the duty operating pumps in the event of high peroxide demand	2
	Dosing Method				Duty/duty/duty/standby pumps	Duty/duty/duty/standby pumps	Duty/duty/duty/standby pumps	Duty/duty/duty/standby pumps	1	Northern Star	35		2
	Dosing Control Method				Online CN _{NAD} Analyser	Online CN _{NAD} Analyser	Online CN _{NAD} Analyser	Online CN _{NAD} Analyser	1	Northern Star	35		A
	19.0 REAGENTS												
19.1	Activator - Copper Sulphate												
	Delivery Form		L		Bulk Liquid	Bulk Liquid	Bulk Liquid	Bulk Liquid	1	Northern Star	64		A
	Delivery Size per Trailer				24,000	24,000	24,000	24,000	5	Vendor	112		1
	Number of Trailers per delivery	Maximum			2	2	2	2	1	Northern Star	64		A
	Delivery Size		L		48,000	48,000	48,000	48,000	6	Calculated			
	Delivery System				Compressed Air Displacement	Compressed Air Displacement	Compressed Air Displacement	Compressed Air Displacement	1	Northern Star	64		A
	Delivery Side				Central Rear	Central Rear	Central Rear	Central Rear	1	Northern Star	64		B
	Solution Mix Strength (as CuSO ₄ .5H ₂ O)		% w/v		25	25	25	25	1	Northern Star	16		A
	Solution Mix Strength (as CuSO ₄)		% w/v		16	16	16	16	1	Northern Star	64		A
	Supply Solution SG		t/m³		1.17	1.17	1.17	1.17	1	Northern Star	16		A
	Storage Volume	Minimum	# deliveries		2.5	2.5	2.5	2.5	1	Northern Star	83		B
	Number of Storage Tanks				1	1	1	1	1	Northern Star	16		A
	Live Volume per Tank		m³		120	120	120	120	6	Calculated			B
	Residence Time	Actual	d		11	12	11	10	6	Calculated		Residence time calculated using nominal consumption rates.	
	Consumption												
	SAG Mill Discharge	Nominal	g/t fresh feed to Cyclone Feed Hopper		35	35	35	35	1	Northern Star	23, 75	Added at Cyclone Feed Hopper	A
	Cleaner Scavenger	Nominal	g/t fresh cleaner scav feed		10	10	10	10	1	Northern Star	23, 75		A
	SAG Mill Discharge	Design	g/t fresh feed to Cyclone Feed Hopper		70	70	70	70	6	Calculated		Added at Cyclone Feed Hopper. 100% Design Factor for pumping.	
	Cleaner Scavenger	Design	g/t fresh cleaner scav feed		20	20	20	20	6	Calculated		100% Design Factor for pumping.	
	SAG Mill Discharge	Nominal	L/h		440	414	440	489	6	Calculated			
	Cleaner Scavenger	Nominal	L/h		7.3	5.8	6.3	8.1	6	Calculated			
	SAG Mill Discharge	Design	L/h		881	829	881	978	6	Calculated		100% Design Factor for pumping.	
	Cleaner Scavenger	Design	L/h		14.7	11.6	12.7	16.3	6	Calculated		100% Design Factor for pumping.	
	Nominal Consumption	as CuSO ₄ .5H ₂ O	L/h		447	414	440	489	6	Calculated			
		as CuSO ₄ .5H ₂ O	m³/h		0.45	0.41	0.44	0.49	6	Calculated			
	Design Consumption	as CuSO ₄ .5H ₂ O	L/h		896	841	894	995	6	Calculated			
		as CuSO ₄ .5H ₂ O	m³/h		0.90	0.84	0.89	0.99	6	Calculated			
19.2	Collector - PAX												
	Delivery Form		L		Bulk Liquid	Bulk Liquid	Bulk Liquid	Bulk Liquid	1	Northern Star	64	Potassium Amyl Xanthate	A
	Delivery Size per Trailer				24,000	24,000	24,000	24,000	5	Vendor	112		1
	Number of Trailers per delivery	Maximum			2	2	2	2	1	Northern Star	64		A
	Delivery Size		L		48,000	48,000	48,000	48,000	6	Calculated			
	Delivery System				Compressed Air Displacement	Compressed Air Displacement	Compressed Air Displacement	Compressed Air Displacement	1	Northern Star	16		A
	Delivery Side				Central Rear	Central Rear	Central Rear	Central Rear	1	Northern Star	64		B
	Solution Mix Strength		% w/v		20	20	20	20	1	Northern Star	64		A
	Supply Solution SG		t/m³		1.095	1.095	1.095	1.095	1	Northern Star	23		A
	Number of Storage Tanks				1	1	1	1	1	Northern Star	16		A
	Live Volume per Tank		m³		270	270	270	270	6	Calculated			B
	Residence Time	Actual	d		7	7	7	6	2	Primero	10	Residence time calculated using nominal consumption rates.	A
		Required	d		7	7	7	7	1	Northern Star	10, 16		A
	Consumption												
	Rougher Feed	Nominal	g/t fresh mill feed		40	40	40	40	1	Northern Star	16, 75		A
	Scavenger Feed	Nominal	g/t fresh mill feed		35	35	35	35	1	Northern Star	16, 75		A
	Cleaner Conditioning	Nominal	g/t fresh mill feed		16	16	16	16	1	Northern Star	16, 75		A
	Cleaner Scavenger	Nominal	g/t fresh mill feed		16	16	16	16	1	Northern Star	16, 75		A
	Rougher Feed	Nominal	L/h		629	592	629	699	6	Calculated			
	Scavenger Feed	Nominal	L/h		551	518	551	611	6	Calculated			

AREA	PARAMETER	UNIT	DATA				REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV
			1	2	3	4					
			FIMISTON	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um					
19.3	Frother - Luprofroth 168										
	Delivery Form			Bulk Liquid	Bulk Liquid	Bulk Liquid	Bulk Liquid	1	Northern Star	64	A
	Delivery Size per Trailer			26,000	26,000	26,000	26,000	5	Vendor		0
	Number of Trailers per delivery	Maximum		1	1	1	1	1	Northern Star	16	A
	Delivery Size	L		26,000	26,000	26,000	26,000	6	Calculated		
	Delivery System			Pumped Transfer From Delivery Truck	Pumped Transfer From Delivery Truck	Pumped Transfer From Delivery Truck	Pumped Transfer From Delivery Truck	1	Northern Star	64	A
	Delivery Side			Passenger	Passenger	Passenger	Passenger	1	Northern Star	64	B
	Solution Mix Strength	% w/w		100	100	100	100	1	Northern Star	64	A
	Supply Solution SG	tm³		1.082	1.082	1.082	1.082	1	Northern Star	16	A
	Number of Storage Tanks			1	1	1	1	2	Primero		A
	Live Volume per Tank	m³		60	60	60	60	1	Northern Star	16	A
	Residence Time	Minimum Actual	d d	7 29	7 30	7 29	7 26	1 6	Northern Star Calculated	10	A
	Consumption										
	Rougher Feed	Nominal	g/t fresh mill feed	12	12	12	12	1	Northern Star	16, 75	A
	Scavenger Feed	Nominal	g/t fresh mill feed	6	6	6	6	1	Northern Star	16, 75	A
	Cleaner Feed	Nominal	g/t fresh mill feed	6	6	6	6	1	Northern Star	44, 75	A
	Cleaner Scavenger	Nominal	g/t fresh mill feed	6	6	6	6	1	Northern Star	16, 75	A
	Rougher Feed	Nominal	L/h	34.9	32.9	34.9	38.8	6	Calculated		
	Scavenger Feed	Nominal	L/h	17.4	16.4	17.4	19.4	6	Calculated		
	Cleaner Feed	Nominal	L/h	17.4	16.4	17.4	19.4	6	Calculated		
	Cleaner Scavenger	Nominal	L/h	17.4	16.4	17.4	19.4	6	Calculated		
	Total Consumption	Nominal	L/h	87.2	82.1	87.2	96.9	6	Calculated		
			m³/h	0.09	0.08	0.09	0.10	6	Calculated		
	Consumption										
	Rougher Feed	Design	g/t fresh mill feed	24	24	24	24	6	Calculated		
	Scavenger Feed	Design	g/t fresh mill feed	12	12	12	12	6	Calculated		
	Cleaner Feed	Design	g/t fresh mill feed	12	12	12	12	6	Calculated		
	Cleaner Scavenger	Design	g/t fresh mill feed	12	12	12	12	6	Calculated		
	Rougher Feed	Design	L/h	69.8	65.7	69.8	77.5	6	Calculated		
	Scavenger Feed	Design	L/h	34.9	32.9	34.9	38.8	6	Calculated		
	Cleaner Feed	Design	L/h	34.9	32.9	34.9	38.8	6	Calculated		
	Cleaner Scavenger	Design	L/h	34.9	32.9	34.9	38.8	6	Calculated		
	Total Consumption	Design	L/h	174.4	164.3	174.4	193.8	6	Calculated		
			m³/h	0.17	0.16	0.17	0.19	6	Calculated		
19.4	Flocculant										
	Packaging Type			Bulk Powder	Bulk Powder	Bulk Powder	Bulk Powder	1	Northern Star	64	A
	Delivery Size per Trailer			22,500	22,500	22,500	22,500	1	Northern Star	64	A
	Number of Trailers per delivery	Maximum		1	1	1	1	1	Northern Star	64	A
	Delivery Size	kg		22,500	22,500	22,500	22,500	6	Calculated		
	Delivery System			Pneumatic Compressor on Bulk Tanker	Pneumatic Compressor on Bulk Tanker	Pneumatic Compressor on Bulk Tanker	Pneumatic Compressor on Bulk Tanker	1	Northern Star	64	C
	Delivery Side			Passenger	Passenger	Passenger	Passenger	1	Northern Star	64	B
	Storage Volume	Minimum	# deliveries	2	2	2	2	1	Northern Star	23	A
	Number of Mixing Tanks			1	1	1	1	1	Northern Star	16	A
	Mixing Tank Live Volume		m³	40	40	40	40	5	Vendor	36	B
	Solution Mix Strength	% w/v		0.30	0.30	0.30	0.30	1	Northern Star	44	A
	Mixes per Day per Tank			14	14	15	16	6	Calculated		
	Flocculant Dilution Factor			10	10	10	10	1	Northern Star	103	B
	Number of Storage Tanks			1	1	1	1	5	Vendor	36	A
	Storage Tank Live Volume		m³	450	450	450	450	5	Vendor	36	A
	Residence Time		h	19	19	18	17	6	Calculated		
	Consumption										
	Flotation Tailings Thickener	Nominal	g/t	20	20	20	20	1	Northern Star	23	A
		Nominal	kg/h	61.4	57.2	60.8	68.2	6	Calculated		
		Nominal	L/h	20,473	19,054	20,738	22,738	6	Calculated		
		Design	L/h	40,946	38,108	40,557	45,476	6	Calculated		
	Concentrate Pre-Leach Thickener	Nominal	g/t	70	70	70	70	1	Northern Star	23	A
		Nominal	kg/h	5.2	7.3	7.3	5.8	6	Calculated		
		Nominal	L/h	1,748	2,430	2,430	1,949	6	Calculated		
		Design	L/h	3,496	4,859	4,860	3,897	6	Calculated		
	Concentrate Thickener	Nominal	g/t	70	70	70	70	1	Northern Star	23	A
		Nominal	kg/h	5.2	7.3	7.3	5.8	6	Calculated		
		Nominal	L/h	1,748	2,430	2,430	1,948	6	Calculated		
		Design	L/h	3,496	4,859	4,860	3,897	6	Calculated		
	Total Nominal Consumption	Stage 2	g/h	71,909	71,742	75,417	79,908	6	Calculated		
		Stage 2	L/h	23,970	23,914	25,139	26,636	6	Calculated		
		Stage 2	m³/h	24.0	23.9	25.1	26.6	6	Calculated		
		Stage 2	t/a	588	587	617	654	6	Calculated		
	Total Design Consumption	Stage 2	g/h	95,000	95,000	95,000	95,000	2	Primero		A
		Stage 2	t/a	777	777	777	777	6	Calculated		
		Stage 2	m³/h	31.7	31.7	31.7	31.7	6	Calculated		
		Stage 2	L/h	47,938	47,826	50,277	53,270	6	Calculated		
		Stage 2	m³/h	47.9	47.8	50.3	53.3	6	Calculated		
	Silo Capacity	Actual	t d	45 26	45 26	45 25	45 23	5 6	Vendor Calculated	36	A
19.6	pH Modifier - Quicklime										
	Type			New Powder	New Powder	New Powder	New Powder				
	Packaging Type			Bulk	Bulk	Bulk	Bulk	1	Northern Star	64	B
	Delivery Size per Trailer		kg	30,000	30,000	30,000	30,000	1	Northern Star	64	A
	Number of Trailers Per Delivery	#		3	3	3	3	1	Northern Star	64	A
	Delivery Size	kg		90,000	90,000	90,000	90,000	6	Calculated		
	Preferred Delivery Method			Pneumatic Compressor on Bulk Tanker	Pneumatic Compressor on Bulk Tanker	Pneumatic Compressor on Bulk Tanker	Pneumatic Compressor on Bulk Tanker	1	Northern Star	64	C
	Bulk Density		tm³	0.95	0.95	0.95	0.95	1	Northern Star	16	A
	Solids SG			3.37	3.37	3.37	3.37	1	Northern Star	16	A
	Water SG			1.07	1.07	1.07	1.07	1	Northern Star	16	A
	Solution Mix Strength	% w/w		20	20	20	20	1	Northern Star	16	A
	Lime Slurry Density	tm³		1.24	1.24	1.24	1.24	6	Calculated		
	Feed Size										
	F100		mm	6.7	6.7	6.7	6.7	1	Northern Star	116	1
	F80		µm	1,150	1,150	1,150	1,150	1	Northern Star	116	1
	Product Size		µm	38	38	38	38	1	Northern Star	116	1
	Dry Consumption										
	Concentrate CIL - Nominal	concentrate basis	kg/t	51.2	51.2	51.2	51.2	1	Northern Star	23	A
	Concentrate CIL - Design	concentrate basis	kg/t	120.1	120.1	120.1	120.1	1	Northern Star	16	A
	Flotation Tailings CIL - Nominal	tailings CIL basis	kg/t	3.20	3.20	3.20	3.20	1	Northern Star	23	A
	Flotation Tailings CIL - Design	tailings CIL basis	kg/t	4.50	4.50	4.50	4.50	1	Northern Star	16	A
	Total Dry Consumption										
	Nominal	@ 93.4% Availability	th	13.9	14.8	15.4	15.6	6	Calculated		
	Nominal	@ 75% Lime Utilisation	th	17.3	18.5	19.2	19.3	6	Calculated		
	Design	@ 93.4% Availability	th	23.2	25.9	26.7	25.8	6	Calculated		
	Design	@ 75% Lime Utilisation	th	28.9	32.2	33.2	32.1	6	Calculated		
	Milk of Lime Consumption										
	Nominal		m³/h	56.2	59.9	62.2	62.4	6	Calculated		
	Peak		m³/h	93.5	104.4	107.7	104.0	6	Calculated		
	New Circuit										
	Storage Silo Capacity	per Silo	t	1,132	1,132	1,132	1,132	5	Vendor	118	1
	Number of Silos			2	2	2	2	5	Vendor	118	1
	Residence Time	Actual	d	7	6	6	6	6	Calculated		
	Lime Slaker Type			MIII	MIII	MIII	MIII	1	Northern Star	43	A
	Number of Slakers			2	2	2	2	5	Vendor	118	1
	Slaker Power		kW/mill	6 x 55	6 x 55	6 x 55	6 x 55	5	Vendor	118	2
	Utilisation	%		75	75	75	75	1	Northern Star	116	A
	Mill Throughput	Design	thr/mill	17	17	17	17	5	Vendor	118	2
			t/hr	33	33	33	33	5	Vendor	118	2
	Make-up Water Type			Raw Water	Raw Water	Raw Water	Raw Water	1	Northern Star	103	A
	Milk of Lime Storage Tank Live Volume	Max	m³	524	524	524	524	1	Northern Star	107	0
	Number of Storage Tanks			1	1	1	1	1	Northern Star	43	A
	Residence Time	Actual	h	9.3	8.8	8.4	8.4	6	Calculated		
	Ringmain Flow Factor			2.5	2.5	2.5	2.5	2	Primero		C
	Lime Ringmain Pumps	Required	m³/h	234	261	269	260	6	Calculated		
		Selected	m³/h	300	300	300	300	2	Primero		C

AREA	PARAMETER	UNIT	DATA				REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV
			1	2	3	4					
			FIMISTON	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um					
				NEW CIRCUIT 28.35 MTPA THROUGHPUT NOMINAL GRADE P80 220um							
19.7	Hydrogen Peroxide										
	Type			Liquid	Liquid	Liquid	Liquid	1	Northern Star	64	A
	Supply Strength	% w/v		70	70	70	70	1	Northern Star	64	B
	Supply Solution SG	tim³		1.29	1.29	1.29	1.29	5	Vendor	111	1
	pH			4.3	4.3	4.3	4.3	1	Northern Star	44	B
	Delivery Form			Bulk, Isolainer	Bulk, Isolainer	Bulk, Isolainer	Bulk, Isolainer	1	Northern Star	64	B
	Delivery Size per Trailer	L		24,000	24,000	24,000	24,000	5	Vendor	64	B
	Number of Trailers Per Delivery	#		2	2	2	2	1	Northern Star	64	0
	Delivery Size	L		48,000	48,000	48,000	48,000	6	Calculated		
	Delivery System			Compressed Air Displacement	Compressed Air Displacement	Compressed Air Displacement	Compressed Air Displacement	1	Northern Star	64	B
	Delivery Side			Central Top	Central Top	Central Top	Central Top	1	Northern Star	64	B
	No. Dosing Points			3	3	3	3	1			
	Consumption Rate - Nominal	each Inline Mixer	L/h	140	140	140	140	1	Northern Star	35	One dosing pump per mixer.
	Consumption Rate - Design	each Inline Mixer	L/h	420	420	420	420	6	Calculated		
		Total	L/h	280	280	280	280	6	Calculated		
		Total	L/h	840	840	840	840	6	Calculated		
	Number of Storage Tanks			1	1	1	1	2	Primero		316 SS Storage Tank
	Storage Tank Live Volume	Per Tank	m³	84.35	84.35	84.35	84.35	2	Primero		2000-F72-TQY-ME-10322_1
		Total	m³	84	84	84	84	6	Calculated		
	Hydrogen Peroxide Storage Tank Residence Time			7	7	7	7	1			
	Required Residence Time	Nominal	days	8	8	8	8	1	Northern Star	10	
	Nominal Usage Rate		days					6	Calculated		
19.8	Sodium Cyanide										
	Delivery Form			New Bulk Liquid	New Bulk Liquid	New Bulk Liquid	New Bulk Liquid	1	Northern Star	64	B
	Delivery Size per Trailer	L		22,000	22,000	22,000	22,000	1	Northern Star	64	A
	Number of Trailers per delivery	Maximum		2	2	2	2	1	Northern Star	64	A
	Delivery Size	L		44,000	44,000	44,000	44,000	6	Calculated		
	Delivery System			Compressed Air Displacement	Compressed Air Displacement	Compressed Air Displacement	Compressed Air Displacement	1	Northern Star	64	C
	Delivery Side			Central Top	Central Top	Central Top	Central Top	1	Northern Star	64, 83	One per Storage Tank (2 off)
	Solution Mix Strength	% w/v		29	29	29	29	1	Northern Star	64	
	Supply Solution SG	tim³		1.24	1.24	1.24	1.24	2	Primero		NaCN SG at 29%w/v solution (Source: SysCAD)
	Consumption										
	Flotation Concentrate CIL	Operating Range	kg/t CIL Feed	2.0 - 8.0	2.0 - 8.0	2.0 - 8.0	2.0 - 8.0	1	Northern Star	23, 44	Dosed to six tanks
		No. of Dose Points	#	6	6	6	6	1	Northern Star	23, 44, 103	Dosed to six tanks
		Nominal	kg/t CIL per tank	3.58	3.58	3.58	3.58	1	Northern Star	44	Dosed to six tanks. Total of 21.50 kg/t as per NSR provided data.
		Nominal	kg/t CIL Total	21.5	21.5	21.5	21.5	6	Calculated		
		Nominal	kg/d	38,658	33,733	33,740	43,093	6	Calculated		
	Flotation Concentrate CIL	Design	kg/t CIL per tank	10	10	10	10	1	Northern Star	23, 44	Dosed to six tanks
		Design	kg/t CIL Total	60	60	60	60	6	Calculated		
		Design	kg/d	107,882	149,951	149,973	120,260	6	Calculated		of NaCN
	Flotation Tailings CIL	Operating Range	kg/t CIL Feed	0.04-0.24	0.04-0.24	0.04-0.24	0.04-0.24	1	Northern Star	23	
		No. of Dose Points	#	4	4	4	4	1	Northern Star	103	CIL4 Feed Distribution Box, Leach tank 1, 2 and Adsorption Tank 1
	Flotation Tailings CIL	Nominal	kg/t CIL Total	0.22	0.22	0.22	0.22	1	Northern Star	23	Dosing split between dosing points
		Nominal	kg/d	16,637	16,678	16,648	18,480	6	Calculated		
	Flotation Tailings CIL	Design	kg/t CIL Feed	0.33	0.33	0.33	0.33	1	Northern Star	16	Dosing split between dosing points
		Design	kg/d	24,956	23,517	24,972	27,720	6	Calculated		
	Elution Circuit 1	CIL2/3	m³/strip	1.2	1.2	1.2	1.2	1	Northern Star	16	
		Min. instantaneous	m³/h	2.4	2.4	2.4	2.4	6	Calculated		Assumed 30 min pre-treatment time
	Elution Circuit 2	CIL2/3	m³/strip	1.2	1.2	1.2	1.2	1	Northern Star	16	
		Min. instantaneous	m³/h	2.4	2.4	2.4	2.4	6	Calculated		Assumed 30 min pre-treatment time
	Elution Circuit 3	CIL4	m³/strip	1.51	1.51	1.51	1.51	6	Calculated		
		Min. instantaneous	m³/h	3.03	3.03	3.03	3.03	6	Calculated		
	Elution Consumption	Nominal	kg/d	2,375	2,375	2,375	2,375	6	Calculated		
		Design	kg/d	4,750	4,750	4,750	4,750	6	Calculated		100% Design Factor
	Intensive Leach		kg/batch	230	230	230	230	5	Vendor		
		Nominal	kg/d	230	230	230	230	6	Calculated	71	Based on CS10000 Acacia unit
		Design	kg/d	460	460	460	460	6	Calculated		100% Design Factor
	Nominal Consumption		kg/d	57,900	72,016	72,993	64,178	6	Calculated		
			kg/h	2,413	3,001	3,041	2,674	6	Calculated		
		NaCN solution (29%w/v)	L/h	6,734	8,375	8,489	7,464	6	Calculated		
			kg/t plant feed	0.77	1.01	0.97	0.77	6	Calculated		
	Design Consumption		kg/d	138,048	178,678	180,155	153,190	6	Calculated		
			kg/h	5,752	7,445	7,506	6,383	6	Calculated		
		NaCN solution (29%w/v)	L/h	16,054	20,779	20,951	17,815	6	Calculated		
			kg/t plant feed	1.83	2.51	2.39	1.83	6	Calculated		
	New Circuit										
	Number of Storage Tanks			2	2	2	2	1	Northern Star	83	
	Live Volume per Tank	Each Tank	m³	1,250	1,250	1,250	1,250	1	Northern Star	83	
	Residence Time	Actual	d	15.5	12.4	12.3	14.0	6	Calculated		Based on nominal consumption.
	Ringmain Flow Factor			2.0	2.0	2.0	2.0	2	Primero		Applied to Design flow consumption rates
	Cyanide Ringmain Pumps	Required	m³/h	32.1	41.6	41.9	35.6	6	Calculated		
		Selected	m³/h	50	50	50	50	2	Primero		
19.9	Oxygen			New							
	Flotation Tails CIL				0.17			6	Calculated	91	NSR Supplied Package
	Consumption Rate	Nominal	Nm³/t mill feed								
	Flotation Concentrate CIL				13			1	Northern Star	23	
	Consumption Rate	Nominal	Nm³/h conc								
	Total Consumption										
		Nominal	t/d		53.2			6	Calculated	91	
		Maximum	t/d		70.8			6	Calculated	91	
	New Oxygen Plant				70						
	VPSA Capacity	Selected	t/d					1	Northern Star	47	NSR Supplied Package
19.10	Lead Nitrate										
	Delivery Form			New Bulk Liquid	New Bulk Liquid	New Bulk Liquid	New Bulk Liquid	1	Northern Star	64	A
	Delivery Size per Trailer	L		24,000	24,000	24,000	24,000	5	Vendor	112	1
	Number of Trailers per delivery	Maximum		2	2	2	2	1	Northern Star	64	A
	Delivery Size	L		48,000	48,000	48,000	48,000	6	Calculated		
	Preferred Delivery System			Compressed Air Displacement	Compressed Air Displacement	Compressed Air Displacement	Compressed Air Displacement	1	Northern Star	64	A
	Delivery Side			Central Rear	Central Rear	Central Rear	Central Rear	1	Northern Star	64	B
	Solution Mix Strength	% w/v		40	40	40	40	1	Northern Star	64	A
	Supply Solution SG	tim³		1.17	1.17	1.17	1.17	1	Northern Star	16	A
	Consumption										
19.11	Sodium Hydroxide (Caustic)										
	Delivery Form			New Bulk Liquid				1	Northern Star	64	A
	Delivery Size per Trailer	L		20,000				1	Northern Star	64	A
	Number of Trailers per delivery	Maximum		3				1	Northern Star	64	A
	Delivery Size	L		60,000				6	Calculated		
	Preferred Delivery System			Pumped Transfer From Truck				1	Northern Star	64	B
	Delivery Side			Passenger				1	Northern Star	64	B
	Solution Mix Strength	% w/v		50				1	Northern Star	64	A
	Supply Solution SG	tim³		1.50				1	Northern Star	16	A
	Consumption										
	Elution Circuit 1		m³/strip		1.7			1	Northern Star	16	Existing
		Min. instantaneous	m³/h		3.4			6	Calculated		Assumed 30 min pre-treatment time
	Elution Circuit 2		m³/strip		1.7			1	Northern Star	16	
		Min. instantaneous	m³/h		3.4			6	Calculated		Assumed 30 min pre-treatment time
	Elution Circuit 3		m³/strip		0.85			6	Calculated		
		Min. instantaneous	m³/h		1.70			6	Calculated		
	Intensive Leach		kg/batch		29			5	Vendor	71	Based on CS10000 Acacia unit
			L/batch		39			6	Calculated		
	Consumption										
	Elution Circuit 1		kg/d		1,913			6	Calculated		
	Elution Circuit 2		kg/d		1,913			6	Calculated		
	Elution Circuit 3		kg/d		1,277			6	Calculated		
	Intensive Leach		kg/d		58			6	Calculated		Based on two batches per day
	Calculated Consumption										
		Nominal	kg/d		5,160			6	Calculated		
		Nominal	kg/h		215			6	Calculated		
		Nominal	L/h		287			6	Calculated		
	New Circuit										
	Number of Storage Tanks				1			2	Primero		C
	Live Volume		m³		250			2	Primero		A
	Residence Time	Nominal	d		36			6	Calculated		Based on Nominal Consumption
	Ringmain Flow Factor				2.5			2	Primero		Applied to combined instantaneous flowrate from Elution Circuits 1 & 2
	Caustic Ringmain Pumps	Required	m³/h		17			6	Calculated		
		Selected	m³/h		20			2	Primero		

AREA	PARAMETER	UNIT	DATA				REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV
			1	2	3	4					
			FIMISTON	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um					
19.12	Hydrochloric Acid			New	Bulk Liquid						
	Delivery Form			20,000			1	Northern Star	64		A
	Delivery Size per Trailer			3			1	Northern Star	64		A
	Number of Trailers per delivery	Maximum		60,000			1	Northern Star	64		A
	Delivery Size	L		Pumped Transfer From Truck			6	Calculated			A
	Preferred Delivery System			Passenger			1	Northern Star	64		A
	Delivery Side			32			1	Northern Star	64		B
	Supply Strength	% w/w		1.16			1	Northern Star	64		A
	Supply Solution SG	t/m³					1	Northern Star	16		A
	Consumption										
	Acid Wash Column 1	m³/strip		2.7			1	Northern Star	16		A
		Min. instantaneous		9.5			6	Calculated			
	Acid Wash Column 2	m³/strip		2.7			1	Northern Star	16		A
		Min. instantaneous		9.5			6	Calculated			
	Acid Wash Column CIL4	m³/strip		1.5			6	Calculated			
		Min. instantaneous		5.2			6	Calculated			
	Consumption										
	Acid Wash Column 1	kg/d		1,503			6	Calculated			
	Acid Wash Column 2	kg/d		1,503			6	Calculated			
	Acid Wash Column CIL4	kg/d		1,085			6	Calculated			
	Calculated Consumption										
		kg/d		4,092			6	Calculated			
		kg/h		170			6	Calculated			
		L/h		256			6	Calculated			
		L/h		459			6	Calculated		MB estimated, based on CIL4 acid injection time (17 min) Based on 2.7 m³ HCl / strip	
	New Circuit										
	Number of Storage Tanks			1			1	Northern Star	24		A
	Live Volume per Tank	m²		200			2	Primero			A
	Residence Time	Nominal		18			6	Calculated		Based on Nominal Consumption	
	Dilute Acid Storage Tank Live Volume	m²		25			2	Primero	110		0
	Neat Acid Dosing Pumps	Required		20			2	Primero		2 off Elution Circuit 1/2 strips occurring simultaneously	C
	Dilute Acid Dosing Pump	Minimum		64			2	Primero			0
19.13	Grinding Media										
	SAG Mill										
	Number of Bunkers			2			1	Northern Star	16		A
	Bunker Storage Size	m³		35			1	Northern Star	16		A
	Media Type			Forged Steel			1	Northern Star	16, 38		A
	Media Diameter	mm		125			1	Northern Star	16, 38		A
	Bulk Density	t/m³		4.65			1	Northern Star	16		A
	Package Form			Bulk			1	Northern Star	16		A
	Delivery Size	t		25			1	Northern Star	23		A
	Trailers per Delivery	#		1			1	Northern Star	23		A
	Consumption Rate - Existing SAG	kg/t mill feed		0.35			1	Northern Star	16, 38		A
	Consumption Rate - New SAG	kg/t mill feed		0.40			1	Northern Star	16, 38		A
	Total Consumption	t/a		9,759			6	Calculated			
	Storage Capacity	t		325.5			6	Calculated			
	Residence Time	d		12			6	Calculated			
	Indexer/Charger Storage Capacity	Minimum		1.5			1	Northern Star	16	CAT972 Bucket	A
	Ball Mill										
	Number of Bunkers			1			1	Northern Star	16		A
	Bunker Storage Size	m³		35			1	Northern Star	16		A
	Media Type			High Chrome			1	Northern Star	16, 38		A
	Media Diameter	mm		75			1	Northern Star	16, 38		A
	Bulk Density	t/m³		4.75			1	Northern Star	16		A
	Package Form			Bulk			1	Northern Star	16		A
	Delivery Size	t		25			1	Northern Star	23		A
	Trailers per Delivery	#		1			1	Northern Star	23		A
	Consumption Rate	kg/t mill feed		0.20			1	Northern Star	16, 38		A
	Consumption	t/a		5,148			6	Calculated			
	Storage Capacity	t		166.3			6	Calculated			
	Residence Time	d		12			6	Calculated			
	Indexer/Charger Storage Capacity	Minimum		1.5			1	Northern Star	16	CAT972 Bucket	A
	UFG Mill										
	Media Type			Ceramic			1	Northern Star	16		A
	Media Diameter	mm		2.5			1	Northern Star	16		A
	Bulk Density	t/m³		2.2			1	Northern Star	16		A
	Package Form			Bulk Bag			1	Northern Star	16		A
	Package Weight	kg		2,000			1	Northern Star	16		A
	Addition Rate	kg/t mill feed		0.56			1	Northern Star	10		B
		UFG Mill 1		0.0			6	Calculated			
		UFG Mill 1		0.0			6	Calculated			
		UFG Mill 2		21.0			6	Calculated			
		UFG Mill 2		503.4			6	Calculated			
		UFG Mill 3		21.0			6	Calculated			
		UFG Mill 3		503.4			6	Calculated			
	Site Storage Capacity	Minimum		50			1	Northern Star	10		B
		Design		54			2	Primero		Navisworks model. Designed to achieve the minimum.	
		days		107			6	Calculated		UFG Mill 1, 2 & 3 Consumption	B
	Lime Slaking Mills										
	Media Type			Steel			5	Vendor	118		2
	Media Diameter	mm		25.0			5	Vendor	118		2
	Bulk Density	t/m³		4.85			5	Vendor	118		2
	Total Consumption	kg/mill/week		170			5	Vendor	118	Estimated. Dependant on lime slaking usage and hardness	2
		kg/week total		340			6	Calculated		Estimated using vendor provided mill ball consumption and lime design throughput	
		kg/t		10.2			6	Calculated			
	Package Form			Bulk Bag			1	Northern Star	118		2
	Media Hopper Capacity	UFG Mill 2		14			5	Vendor	65	Isacharger capacity negligible.	B
	Residence Time	UFG Mill 2		61.2			6	Calculated			
	Residence Time	# Bags		15.4			6	Calculated			
	Media Hopper Capacity	UFG Mill 3		14			5	Vendor	65	Isacharger capacity negligible.	B
	Residence Time	UFG Mill 3		61.2			6	Calculated			
	Residence Time	# Bags		15.4			6	Calculated			
19.14	Antiscalant										
	Type			FF585			1	Northern Star	16	Dosing pump for each dosing point	A
	Addition Rate - Gravity Concentrators	ml/min		40			1	Northern Star	16	New addition in Stage 1, to be dosed into gravity concentrator fluidisation water line	A
	Addition Rate - Kiln Scrubber (Existing)	ml/min		45			1	Northern Star	16	New addition in Stage 2	A
	Addition Rate - CIL4 Kiln Mercury Abatement System	ml/min		61			1	Northern Star	44	New addition in Stage 2	B
	Total Consumption	ml/min		146			6	Calculated			
	Number of IBC's for Storage	No.		2			6	Calculated		Based on 7 days storage & 1,000 L IBC volumes	
	Type			FF540			1	Northern Star	16	Dosing pump for each dosing point	A
	Addition Rate - Elution Circuit 1 (Existing)	ml/min		56			1	Northern Star	16	New addition in Stage 2	A
	Addition Rate - Elution Circuit 2 (Existing)	ml/min		70			1	Northern Star	16	New addition in Stage 2	A
19.15	LPG										
	Type			Propane			1	Northern Star	16		A
	Elution Burner 1 Range	High Flame		4,900			1	Northern Star	16	Existing equipment fed from upgraded LPG Storage Vessel	A
		Low Flame		2,000			1	Northern Star	16	Existing equipment fed from upgraded LPG Storage Vessel	A
	Elution Burner 2 Range	High Flame		4,900			1	Northern Star	16	Existing equipment fed from upgraded LPG Storage Vessel	A
		Low Flame		2,000			1	Northern Star	16	Existing equipment fed from upgraded LPG Storage Vessel	A
	CIL4 Elution Burner Range	Pre-Heat		823			5	Vendor	81	New addition in Stage 2 fed from upgraded LPG Storage Vessel	A
		Steady State		708			5	Vendor	81	New addition in Stage 2 fed from upgraded LPG Storage Vessel	A
	Regeneration Kiln 1	MJ/h		4,500			1	Northern Star	16	Existing equipment fed from upgraded LPG Storage Vessel	A
	Regeneration Kiln 2	MJ/h		4,500			1	Northern Star	16	Existing equipment fed from upgraded LPG Storage Vessel	A
	Regeneration Kiln 3	MJ/h		4,500			1	Northern Star	16	Existing equipment fed from upgraded LPG Storage Vessel	A
	Regenerative Thermal Oxidiser	MJ/h		1,580			1	Northern Star	88	Existing equipment fed from upgraded LPG Storage Vessel	B
	Smelting Furnace	Maximum		2,000			1	Northern Star	16	Existing equipment fed from dedicated Goldroom LPG Storage Vessels	A
	Consumption Rate	Plant Feed Basis		0.40			6	Calculated	50	Inclusive of Regeneration Kiln 3 and RTO but not the Goldroom Smelting Furnace. Calculated from a 25.5 Mtpa basis.	B
	Number of LPG Storage Vessels (Upgraded)			1			1	Northern Star	88	Client supply	B
	Vessel Capacity			161			1	Northern Star	88	Client supply	B
	Residence Time	Actual		5.3			6	Calculated		As per source 82, NSR confirm and accept they won't achieve the 7 days	

AREA	PARAMETER	UNIT	DATA				REF	REFERENCE DESCRIPTION	SOURCE	COMMENTS	REV
			1	2	3	4					
			FIMISTON	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA NOMINAL GRADE P80 180um	NEW CIRCUIT REDUCED THROUGHPUT 24.0 MTPA DESIGN GRADE P80 180um	NEW CIRCUIT NOMINAL THROUGHPUT 25.5 MTPA ACHIEVABLE GRADE P80 180um					
20.0	WATER SERVICES - EXISTING										
20.1	Saline Bore Water										
	Source			Borefields			1	Northern Star	16		A
	Storage Method 1			Dam 1 (North)			1	Northern Star	16		A
	Live Volume	m³		8,000			1	Northern Star	16		A
	Dam Water Pumps			2			1	Northern Star	23		A
	Pump Model	#		Stalker 250-250-400, 132kW			1	Northern Star	23		A
	Storage Method 2			Dam 2 (South)			1	Northern Star	16		A
	Live Volume	m³		12,000			1	Northern Star	16		A
	Dam Water Pumps			3			1	Northern Star	106		O
	Pump Model	#		TKL 250x250 - 400HT			1	Northern Star	23		A
	Total Live Volume	m³		20,000			6	Calculated			
20.2	Mill Circuit Water										
	Source			Concentrate Pre-Leach Thickener O/F Concentrate Thickener O/F	Concentrate Pre-Leach Thickener O/F Concentrate Thickener O/F	Concentrate Pre-Leach Thickener O/F Concentrate Thickener O/F	1	Northern Star			B
	Storage Method			Circuit Water Tank	Circuit Water Tank	Circuit Water Tank	1	Northern Star	16		A
	Live Volume	m³		1,500	1,500	1,500	1	Northern Star	16		A
	Residence Time	min		100	94	94	6	Calculated			
	Water Demand - Nominal	Calculated		896	958	958	6	Calculated			
	Water Demand - Design	m³/h		1,800	1,800	1,800	1	Northern Star	16		A
	Design Flowrate per Pump	m³/h		1,800	1,800	1,800	1	Northern Star	16		A
	Pump Configuration			Duty / Standby	Duty / Standby	Duty / Standby	1	Northern Star	16		A
	Pump Model			APP51-300	APP51-300	APP51-300	1	Northern Star	23		A
20.3	Cyanide Water										
	Source			Decant Return			1	Northern Star	16		A
	Storage Method			Cyanide Water Tank			1	Northern Star	16		A
	Live Volume	m³		1,115			1	Northern Star	16		A
	Cyanide Water Pump Configuration			Duty / Standby			1	Northern Star	16		A
	Cyanide Water Pump Model			TKL 200x200 - 500HT			1	Northern Star	23		A
	Cyanide Transfer Pump Configuration			Duty / Standby			1	Northern Star	16		A
	Cyanide Transfer Pump Model			TKL 100x65 - 315HT			1	Northern Star	23		A
20.4	Scheme Water										
	Water Demand	m³/h		172	172	173	6	Calculated			
	Scheme Water Quality										
	SG	-		1.00	1.00	1.00	1	Northern Star	1		B
	TDS	mg/L		454	454	454	1	Northern Star	1		B
	pH	-		8.0	8.0	8.0	1	Northern Star	1		B
	Chlorides	mg/L	Range	232	232	232	1	Northern Star	1		B
20.5	Fire Water / Safety Shower / Potable Water			By Others			1	Northern Star	69	By Others	B
20.6	Gland Water										
	Source			Dam 2 (South)			1	Northern Star	16		A
	Storage Method			Gland Water Tank			1	Northern Star	16		A
	Live Volume	m³		83			1	Northern Star	16		A
	Pump Configuration			Duty / Standby			1	Northern Star	16		A
	Pump Model			TIK 90kW			1	Northern Star	23		A
20.7	Eastern Borefield Transfer Dam										
	Source			Seepage			1	Northern Star	16		A
	Storage Method			Dam			1	Northern Star	16		A
	Destination			Dam 3			1	Northern Star	79		B
	Live Volume	m³		12,000			1	Northern Star	16		A
	Pump Configuration			Duty / Standby			1	Northern Star	16		A
20.8	Saline Gravity Dam										
	Source			Borefield			1	Northern Star	16		A
	Storage Method			Dam			1	Northern Star	16		A
	Destination			Dam 1 (North)			1	Northern Star	79		B
	Live Volume	m³		16,000			1	Northern Star	16		A
20.9	Kahtais Saline Dam										
	Source			Borefield			1	Northern Star	16		A
	Storage Method			Dam			1	Northern Star	16		A
	Destination			Dam 3			1	Northern Star	79		B
	Live Volume	m³		13,600			1	Northern Star	16		A
	Pump Configuration			Duty / Duty / Standby			1	Northern Star	16		A
21.0	Gidji Saline Transfer Dam										
	Source			Borefield			1	Northern Star	16		A
	Storage Method			Dam			1	Northern Star	16		A
	Destination			Dam 1 (North) and Dam 2 (South)			1	Northern Star	79		B
	Live Volume	m³		340			1	Northern Star	16		A
	Pump Configuration			Duty / Standby			1	Northern Star	16		A
20.10	Southern Surge Tank										
	Storage Method			Southern Surge Tank			1	Northern Star	16		A
	Destination			Dam 2 (South)			1	Northern Star	16		A
	Capacity	m³		575			1	Northern Star	16		A
21.0	WATER SERVICES - NEW										
21.1	Process Water										
	Source			Flotation Tailings Thickener O/F Saline Bore Water Makeup	Flotation Tailings Thickener O/F Saline Bore Water Makeup	Flotation Tailings Thickener O/F Saline Bore Water Makeup	1	Northern Star	10		A
	Storage Method			Tank	Tank	Tank	9	Agreed	10		A
	Live Volume	m³		2,080	2,080	2,080	9	Agreed	10		A
	Residence Time	Minimum	min	15	15	15	1	Northern Star	16		A
	Water Demand	Calculated		5,355	5,152	5,489	6	Calculated			
	Pump Configuration			Duty / Standby	Duty / Standby	Duty / Standby	9	Agreed	10		A
	Process Water Quality										
	SG	-		1.07	1.07	1.07	1	Northern Star	1		B
	TDS	mg/L		140,000	140,000	140,000	1	Northern Star	1		B
	pH	-		4.6	4.6	4.6	1	Northern Star	1		B
	Chlorides	mg/L	Range	20,492 - 86,243	20,492 - 86,243	20,492 - 86,243	1	Northern Star	1		B
21.2	Raw Water										
	Source			Dam 2 (South)	Dam 2 (South)	Dam 2 (South)	1	Northern Star	43	Supplies the Raw Water Tank	A
	Storage Method			Steel Tank	Steel Tank	Steel Tank	1	Northern Star	68		B
	Configuration			Separate Raw, UFG 2/3 Gland, LP and HP Gland	Separate Raw, UFG 2/3 Gland, LP and HP Gland	Separate Raw, UFG 2/3 Gland, LP and HP Gland	1	Northern Star	68		B
	Total Live Volume	Actual	m³	375	375	375	1	Northern Star	68		B
	Total Live Volume	Required	m³	377	377	377	6	Calculated			
	Residence Time	Actual	min	60	60	60	6	Calculated			
	Residence Time	Required	min	60	60	60	9	Agreed	10, 16	Nominal Basis	B
	Total Nominal Water Demand (water consumption)		m³/h	377	377	377	2	Primero	68	Note: Pump design for instantaneous flows in accordance with mechanical calculation.	B
	Pump Configuration			Duty / Standby	Duty / Standby	Duty / Standby	1	Northern Star	16		A
	Source (Greenfields)			Dam 3	Dam 3	Dam 3	1	Northern Star	69	By Others	B
	Live Volume	m³		35,000	35,000	35,000	1	Northern Star	114	By Others	1
21.3	Gland Water										
	Source			Raw Water Tank	Raw Water Tank	Raw Water Tank	1	Northern Star	43		A
	Gland Water Demand	Design	m³/h	92	92	92	2	Primero			O
	Low Pressure Gland Water Pump Configuration			Duty / Standby	Duty / Standby	Duty / Standby	1	Northern Star	43		A
	High Pressure Gland Water Pump Configuration			Duty / Standby	Duty / Standby	Duty / Standby	1	Northern Star	43		A
22.0	AIR SERVICES										
22.1	Low Pressure Flotation Air										
	Number of Flotation Blowers			4			1	Northern Star	49		A
	Operating Configuration			Duty / Duty / Duty / Standby			1	Northern Star	49		A
	Type			Low pressure, oil free screw blower with sound enclosure			1	Northern Star	49		A
	Capacity - Each	Design	FAD m³/h	8,688			1	Northern Star	49		A
	Capacity - Total		FAD m³/h	26,064			6	Calculated			
	Pressure	Design	kPa(g)	119.7			1	Northern Star	49		A
22.2	High Pressure Air										
	Number of Compressors			3			1	Northern Star	43		A
	Operating Configuration			Duty / Duty / Standby			1	Northern Star	43		A
	Type			Oil Injected Rotary Screw Compressor			5	Vendor	45		B
	Capacity - Each	NTP Conditions	L/s	750			5	Vendor	45		A
	Pressure		kPa(g)	980			5	Vendor	45		B
	Duty			New facilities only			2	Primero			A